

Inversion sequences solved by kernel orbits (OEIS A279544, A279567, A279569, A279558)

Square-root asymptotics to every fixed order
and threshold inverses for four Britt–Beaton classes

Three research reports of one external research session
(bundle Reports 118, 119 and 122), merged into one ProveIt report

Parts I–III: 2 October 2026 merged: 5 October 2026

Abstract

An inversion sequence of length n is an integer vector $(e_1, \dots, e_n)$ with $0 \leq e_i < i$. Britt and Beaton (arXiv:2512.21943v3) gave exact generating trees and catalytic functional equations for the classes avoiding a triple of relations, and stated numerically, and explicitly as non-rigorous, laws $a_n \sim C\mu^n n^{-3/2}$ for four classes whose generating functions they believe to be nonalgebraic. This report prints three manuscripts that prove these four laws. Part I (Report 118) treats class 214 (OEIS A279544, $\mu = 4$) and class 1509 (A279567, $\mu = 3 + 2\sqrt{2}$); Part II (Report 119) treats class 1953A (A279569, $\mu = 27/4$), and hence, by the Wilf-equivalence that Britt and Beaton record, class 1953B; Part III (Report 122) treats class 830 (A279558, $\mu = 27/4$). In each case a kernel substitution reduces the catalytic equation to a scalar affine functional equation, which is solved by a normally convergent sum or quotient along contracting Möbius-type orbits seeded at an algebraic kernel root. A global continuation argument excludes competing singularities on the circle of convergence, and an exact rational certificate proves that the square-root amplitude does not vanish. The result is a convergent local Puiseux expansion and coefficient asymptotics to every fixed order, $a_n = C\mu^n n^{-3/2}(1 + c_1/n + \dots + c_K/n^K + O(n^{-K-1}))$, with C , c_1 and c_2 enclosed in certified intervals. Each Part also proves a threshold inverse with a shrinking two-ceiling bracket, centred at a Lambert- W_{-1} root in Parts I and II and expanded in powers of $1/\log M$ with coefficients polynomial in $\log \log M$ in Part III; these inverse theorems are instances of the Lambert-core, reversion and staircase theorems of the repository’s transseries volume. (Dated note, 5 October 2026: corrected after the independent check recorded at the end of Section 30.1. The first wording, “Each Part also proves a Lambert- W_{-1} threshold inverse with a shrinking two-ceiling bracket; these inverse theorems are instances of the Lambert-core and staircase theorems”, is wrong for Part III, whose inverse contains no W_{-1} . Dated note, 7 October 2026: the subtitle had the same error and now omits “Lambert- W_{-1} ”; it read “Square-root asymptotics to every fixed order and Lambert- W_{-1} threshold inverses for four Britt–Beaton classes”.) Václav Kotěšovec’s OEIS amplitudes agree with every certified interval, so no digits are new: the contribution is proof and certification. Nonalgebraicity, D -finiteness, effective inverse constants and the higher corrections remain open. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The three Parts are three manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 612787fb4 (batch 104). All three are dated 2 October 2026 and are printed in the order of the bundle, which is also the order in which they were written:

Part	Report	Delivered title	Classes	Labels
I	118 (base)	Inversion sequence asymptotics from iterated kernels	214 (A279544), 1509 (A279567)	ivk:orb:
II	119	Inversion sequence asymptotics from two kernel roots	1953A (A279569)	ivk:two:
III	122	A279558 asymptotics and inversion	830 (A279558)	ivk:anc:

Each Part prints its manuscript’s text, with every result, proof, remark, question and limitation; statements keep their delivered wording. Sections are numbered continuously and statements within sections; equations are numbered continuously through the report, as in the deliveries. Part I keeps exactly the section, statement and equation numbers of Report 118 (Sections 1–9, equations (1)–(72)). In Part II, Report 119’s Section k is Section $k + 9$, its statement $k.j$ is $(k + 9).j$, and its equation (m) is $(m + 72)$. In Part III, Report 122’s Section k is Section $k + 19$, its statement $k.j$ is $(k + 19).j$, and its equation (m) is $(m + 128)$. No delivered number moved within its Part: the write’s additions are unnumbered notes, two remarks placed last in their sections, and one subsection at the end of each Part.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date. They give cross-references between the Parts, mark the places where two Parts prove the same kind of statement, record checks made at placement, and point to the repository. Each Part ends with a subsection “Further questions and research” (Sections 9.1, 19.1 and 30.1), which collects, under Vladimir’s standing rule of 4 October 2026, every claim of that Part that is stated without proof, with its source, its sketch and what is missing.

What is proved, and where

For each class, a_n is the number of inversion sequences of length n in the class ($a_0 = 1$), μ its growth constant and $\rho = 1/\mu$ its radius of convergence.

Statement	Where
Class 214, A279544, forbidden $e_j \geq e_k$ and $e_i \geq e_k$: radius $1/4$, continuation to a slit disc, unique dominant singularity, all fixed orders with $C > 0$; C to 52 decimal places, c_1, c_2 to 24	Part I, Theorem 1.1, (24), (32)
Class 1509, A279567, forbidden $e_j \geq e_k$ and $e_i > e_k$: radius $3 - 2\sqrt{2}$, global nonvanishing denominator, Δ -domain, all fixed orders; $\widehat{C}$, $\widehat{c}_1$, $\widehat{c}_2$ certified	Part I, Theorem 6.1, (49), (63)
Class 1953A, A279569, forbidden $e_i > e_k$ and $e_j > e_k$: radius $4/27$, determinant gap and removal of boundary poles, Δ -domain, all fixed orders; C to 38 decimal places, d_1, d_2 certified. The counting statements hold also for class 1953B	Part II, Theorem 10.1, Lemma 14.1, Proposition 14.2, (115), (114); [write] Remark 10.2
Class 830, A279558, forbidden $e_i \neq e_j$, $e_j > e_k$, $e_i \geq e_k$: radius $4/27$, zero-free denominator on $ z \leq 149/1000$, Δ -domain, all fixed orders; C to 43 decimal places, d_1, d_2 certified	Part III, Theorem 20.1, Proposition 24.1, Proposition 27.1
Threshold inverses $N(Y) = \min\{n : a_n \geq Y\}$: Lambert- W_{-1} centre, all-order correction recursion, two-ceiling bracket with error $O(u^{-K-1})$ (Parts I and II); centre expanded in powers of $1/L$, $L = \log M$, with coefficients polynomial in $\log L$, error $O((1 + \log L)^{J+1}/L^{J+1})$ (Part III); existential constants. (Dated note, 5 October 2026: Part III's form added after the independent check; the row first read "Lambert- W_{-1} centre, all-order correction recursion, two-ceiling bracket with error $O(u^{-K-1})$ (existential constants)" for all three Parts.)	Part I, Theorem 7.1 (both classes); Part II, Theorem 17.1 and (128); Part III, Theorem 20.2 and Proposition 28.1
Monotonicity $a_{n+1} > a_n$ for $n \geq 1$ by explicit injections	Part I, Section 7 (classes 214, 1509); Part II, Section 17 (1953A); Part III proves $a_{n+1} \geq a_n$ by injection and strictness for large n from the ratio limit (Section 28); [write] Remark 28.2 gives the strict injection

Open in all three Parts

Effective constants for the transfer remainders and the inverse brackets; the algebraic and D -finite status of the four generating functions (Britt and Beaton believe them nonalgebraic); secondary singularities, exponentially small contributions and the maximal continuation of the orbit representations; the correction coefficients beyond the second, their sign pattern and large-order growth; which other classes admit the same method; whether a rigorous proof existed earlier in the literature (not searched beyond Britt–Beaton and the OEIS). See Sections 9.1, 19.1 and 30.1.

Finite computations in all three Parts (brute force, state recurrences, orbit reconstructions, exact interval certificates and mutation campaigns) check models, formulas and implementations; the infinite tails they use are proved in the text, and no asymptotic statement rests on sampling.

The common pipeline

The three manuscripts were written separately and share no passage longer than a sentence of boilerplate (the longest common run of words is 27 words of title-page matter), so every Part is printed in full. What they share is a method, printed here once, with pointers to each class's version:

Step	214 (Part I)	1509 (Part I)	1953A (Part II)	830 (Part III)
Generating tree (Britt–Beaton)	(4)	(34)	(78)	(139)–(142), re-derived with an all-zero correction
Catalytic equation	(6)	(35)–(36)	(79)	(143)–(144)
Kernel map and scalar affine equation	$f_z(x) =$ $zx/(1 - zx)$; (9)	same f_z ; (37), backward (42)	$f_z(x) = 1/(1 - zx)$; (81)	$\phi_z(x) =$ $1 + zx/(1 - zx)^2$; (146)
Algebraic seed or anchor	Catalan root (11)	$X = 1 + zX + zX^2$, (41)	two cubic roots $X(\pm\sqrt{z})$, (85)	$R = 1 + zR^3$, (151)
Formal orbit representation	Proposition 2.1	(45) (a quotient)	Proposition 12.1 (a quotient of two orbits)	Lemma 22.1 (two orbits)
Normal convergence	Lemma 3.1	(48)	Lemma 13.1	Proposition 23.1
No competing singularity	slit disc, $ r < 1$ (Section 3.2)	$ 1 - yV > 1/3$, (49)	determinant gap and removable boundary poles, Lemma 14.1, Proposition 14.2	$ P > e^{-32/3}$, Proposition 24.1
Nonzero amplitude	(21)–(23)	(53)–(56)	(100), (107)	(171), Section 26
Transfer, all fixed orders	(26)–(30)	Theorem 6.1	(112)–(113)	Proposition 27.1
Threshold inverse	Theorem 7.1	Theorem 7.1	Theorem 17.1	Proposition 28.1

Three ingredients recur in the same form, each proved in every Part where it is used:

- *Transfer coefficients.* With $\Gamma(n - \alpha)/\Gamma(n + 1) \sim n^{-\alpha-1} \sum_{j \geq 0} g_j(\alpha) n^{-j}$ and the local expansion $\sum_j h_j t^j$ in $t = \sqrt{1 - z/\rho}$, the relative corrections are $c_1 = \frac{3}{8} - \frac{3}{2}h_3/h_1$ and $c_2 = \frac{25}{128} - \frac{45}{16}h_3/h_1 + \frac{15}{4}h_5/h_1$: (30) with the Bernoulli recursion (29), (113) with (111), and (186)–(187) with (184). The Puiseux coefficients are called $h_j, \hat{h}_j$ in Part I and b_m, b_j in Parts II and III; Part II’s T_k and Part III’s g_k are Part I’s g_j . The transfer itself is the standard theorem of Flajolet and Sedgewick [6, Chapter VI].
- *Certified constants.* Each Part encloses π by Machin’s identity with alternating-series remainders, takes square roots by integer square roots, and rounds every interval operation outward, on decimal lattices from 10^{-60} to 10^{-120} where interval arithmetic is used; the infinite orbit tails are proved bounds, and Taylor coefficients are bounded by Cauchy’s estimate.
- *The inverse.* The four threshold inverses are one theorem: Part I states it for “either sequence”, Part II for A279569 (adding the explicit log log form (128)), and Part III in the coordinates $L = \log M$ with polynomials $P_j(\log L)$ (its (136) is Part II’s (128), see the note there). All three are instances of results of the repository’s transseries volume [9]: the Lambert centre solves $aX + b \log X = \log Y - \log C$ with $a = \log \mu > 0$ and $b = -\frac{3}{2} < 0$, whose large root the branch rule of `p0:thm:lambert-core` assigns to W_{-1} ; the expansion in powers of the centre has the form of the all-orders reversion `p0:thm:lambert-centered`; Part III’s expansion in powers of $1/L$, like Part II’s (128), has the form of the direct polynomial–logarithmic reversion `p0:thm:flattening`, whose remainder $O((1 + H)^{N+1}/L^{N+1})$ is Part III’s factor $(1 + \log L)^{J+1}$, the only rigorous difference from the Lambert-centred form according to `p0:rem:accuracy-claim` (added 5 October 2026, after the independent check); the counting index is, for targets above 1, the integer staircase of `p0:def:three-inverses` with $n_1 = 1$ (all four sequences begin 1, 1, 2), and a two-ceiling bracket whose ceilings agree determines it,

as in the separation condition of `p0:thm:staircase`; and the mean-value step that moves the forward error to the inverse is the estimate of `p0:cor:forward-to-inverse`. The manuscripts bracket the staircase between the inverses of two smooth envelopes $F_K \pm Dx^{-K-1}$ rather than through an admissible interpolation; the conclusion is the same. These are second routes to known mechanics, not new inverse theory.

Notation across the three Parts

No symbol was renamed. Each Part keeps its manuscript's letters, and each Part opens with a short table of its most dangerous readings. The following symbols change meaning between the Parts, or inside one; the text of each Part fixes the meaning there.

Symbol	Meanings
a_n, b_n	Part I: a_n counts class 214 (A279544), b_n counts class 1509 (A279567); in Section 7 a_n “denotes either of these two sequences”. Part I also has the orbit multiplier $a(z, x)$ and forcing term $b(z, x)$ of (10), and $a(u)$, $b(u)$ in Section 4. Parts II and III: a_n counts their own class; Part III's tree labels are (a, b) , with $a = n - h$, $b = h - k$ (138). <i>Tempting false reading:</i> Part I's b_n is a count, its $b(z, x)$ an orbit function.
Puiseux coefficients	Part I: h_m (class 214, (18)) and $\hat{h}_j$ (class 1509, (55)). Part II: b_m (75). Part III: b_j (131).
C, C_j	The amplitude: C and $\hat{C}$ (Part I), $C = C_0$ (Parts II, III). $C_m = Cc_m$ (28) in Part I; C_k , C_j the absolute coefficients in Parts II, III.
Corrections	Relative: $c_j, \hat{c}_j$ (Part I), $d_j = C_j/C$ (Parts II, III). Logarithmic: d_j (Part I, (67)), ℓ_j (Part II), ℓ_k (Part III). Inverse: e_j (Part I), η_j (Part II), polynomials $P_j(\log L)$ (Part III). <i>False reading:</i> Part I's d_j is logarithmic, Parts II–III's d_j relative; and Part I writes $d_j = v_j^{\text{orb}}$, an orbit derivative, inside Section 4 only.
c, d, e as functions	Part I: $c(z, x)$, $d(z, x)$, $e(z, x)$ the class-1509 coefficients (38). Part II: c, d, e of (80) and their orbit values $c_j, e_j, d_j^{\text{orb}}$ (87). Part III: $c(z, x)$, $d(z, x)$ of (147), and e_j an error in (174).
$A, \mathcal{A}, F, \mathcal{F}$	Part I: $A(z)$, $\mathcal{A}(z)$ the generating functions of A279544, A279567; $F(z, x) = B(z, x, 0)$; $\mathcal{F}(z, x)$ the class-1509 quotient (45). Part II: $A(z)$ the generating function; $F(z, v, w)$ the quotient of two seeds (98). Part III: $F(z)$ the generating function, $A = \log C + p \log \lambda$ a constant. <i>False reading:</i> Part III's A is not a generating function.
B, H, G	$B(z, x, y)$ the trivariate tree series in all Parts. H : Part I's local series $H(t)$ (18); Parts II and III: $H(z, x) = B(z, x, 1)$. G : Part I's $G(z, x)$ with $F = 1 + xG$; Part III's orbit quotient $G(z, R)$ (154). Also $B_j(x)$ Bernoulli polynomials (Parts I, II), $B = 1 - y^2 + yU$ (Part I), $B(L)$ (Part II), B_* , B , B_J constants (Part III).
Targets	The threshold is $N(Y)$ (Part I), $N(T)$ (Part II), $N(M)$ with $L = \log M$ (Part III). Part I also has the bound $Y = 177/250$ and the inverse expansion $y(Y)$ (69) beside the series $y(z)$ (40).
L, λ, β, p	Part I: $L = \log \mu$, $\beta = 3/2$. Parts II and III: $\lambda = \log(27/4)$, $p = 3/2$, and $L = \log T$ (Part II, Section 17.2) or $L = \log M$ (Part III). Part II's β is the second fixed point of f_ρ (Section 15). Part III's L_* is a contraction constant.
u, t, y	u : Part I's Lambert centre $u(Y)$ and critical orbit u_j ; Part II's $u = \sqrt{z}$ and Lambert centre $u(T)$; Part III's $R = \frac{3}{2}(1 - u)$ (169) and u_j . t : $\sqrt{1 - 4z}$ and $\sqrt{1 - z/\sigma}$ (Part I), $\sqrt{1 - z/\rho}$ (Parts II, III). y : Part I's $y(z)$, $y(Y)$; Part III's upper orbit y_j and $y = \lambda n$ (190).
ρ, σ, μ	$\rho = 1/4$ (Part I, class 214), $\sigma = 3 - 2\sqrt{2}$ and $\mu = 3 + 2\sqrt{2}$ (class 1509; Part I's Section 7 writes μ for either growth constant), $\rho = 4/27$ (Parts II, III).

Symbol	Meanings
K, M, R	K : constants (Part I), the kernels $K(z, x) = 1 - x + zx + zx^2$ (Part I) and $(x - 1)^2 - zx^3$ (Part II), a truncation order (all), and K_* , K_0 constants (Part III). M : truncation indices and $M = 483/200$ (Part I), $M = 181/100$ (Part II), the target and $M_* = 176/100$ (Part III). R : radius constants (Part I), the orbit limit $R(z, X)$ and $R_* = 149/1000$ (Part II), the ternary series $R(z)$ (Part III).
P, Q, X	P : Part I's kernel polynomial $P(z, x)$ and products P_j ; Part II's bound $P = 2^{230}$; Part III's orbit limit $P(z)$ and inverse polynomials P_j . Q : Part I's $Q(z, x)$, q_M , $q(z, x)$; Part II's orbit limit $Q(z, X)$ and ratio $q = 7/25$; Part III's $Q(z)$, q_j , $q = 1/2$, q_L . X : Part I's root $X(z)$ of $X = 1 + zX + zX^2$; Part II's cubic roots $X_\pm$ and series $X(u)$; Part III's $X = R^2$.
E, D, α	$E_M, E_N, \tilde{E}_N, E_K$ (Part I), $E_N = 2^{244}q^N$, $\mathcal{E}_N$ (Part II), $E_N = 2^{16}q^N$ (Part III): different tail bounds. D : a constant and $D = 1 - yV$ (Part I), D_N a finite determinant and D_K (Part II). α : the exponent in (25) (Part I) and the attracting fixed point (Parts II, III).

Provenance and merge decisions

Three manuscripts of the session bundle of Reports 1–243, all dated 2 October 2026, each titled “Report N ” with no author line; none names an author or a tool, says it is AI-assisted, or carries “prepared for private review” wording, and none pins a ProveIt commit or names a repository path. They arrived in commit 60f54ea06 and were placed by 612787fb4 (batch 104), which split the triage’s eight-source Britt–Beaton cluster into three reports by method (this one, a279571-inversion-cone-walk and a279551-inversion-log-deficit).

- **Part I**, bundle Report 118, archive `Inversion_Sequences_A279544_A279567_Asymptotic_s_and_Inverses_Source.zip` (413,366 bytes, 16 files; `report118.tex`, 1,100 lines, 18 pages). The base: its `report118.tex` became this `article.tex`. Contributes classes 214 and 1509, the generic threshold-inverse theorem, and the forward (class 214) and backward-quotient (class 1509) orbit solutions.
- **Part II**, bundle Report 119, archive `A279569_Two_Kernel_Asymptotics_and_Inverses_Source.zip` (398,990 bytes, 18 files; `report119.tex`, 1,005 lines, 16 pages). It calls itself a self-contained companion to Report 118 and contrasts its continuation argument with Part I’s. Contributes class 1953A, a continuation by real determinant gap, Pringsheim’s theorem and removal of boundary poles, and the explicit log log form of the inverse.
- **Part III**, bundle Report 122, archive `A279558_Asymptotic_Expansion_and_Inverse_Thresholds_Source.zip` (407,593 bytes, 28 files; `report122.tex`, 1,065 lines, 17 pages). Contributes class 830, a re-derived generating tree with the all-zero correction, the ternary anchor and two orbits, a zero-free product on a larger disc, and the inverse in log M coordinates with an all-order discrete localization.

Where the merge had to choose:

1. *Separate report, order and base*. The three share one method and no method with the other five Britt–Beaton manuscripts of the batch, and none of the eight cites another outside this trio; the placement therefore made them their own report. Report 118 is the base (first in the bundle, and the source Report 119 calls its companion); the Parts follow the bundle order.

2. *Overlaps.* No passage is shared verbatim (8-gram overlap at most 3.3% between any two, in short phrases; the longest common run is 27 words of title-page matter), so nothing is printed once by omission: every Part is complete. The shared method is summarized once in “The common pipeline” above, and dated notes mark the places where a Part proves a statement of the same form as another (the inverse theorem, the transfer coefficients, the log log inverse). Part I’s continuation for class 1509 and Part II’s for class 1953A are genuinely different arguments, contrasted by Report 119 itself; both are printed.
3. *Labels, citations and macros.* Every delivered label carries its Part’s prefix; the write added the labels of the three Parts, of the three further-questions subsections, of ten delivered section and subsection headings that had none (listed in the README), of the two write remarks, and of this front matter. Report 119’s citation key `oeis` is `oeis1953` here and Report 122’s is `oeis830`; the three bibliographies are merged, keeping every detail of every entry. The three macro sets are compatible and merged; Report 122’s unused `definition` environment is omitted.
4. *Front matter.* The three title blocks are recorded at the head of their Parts and the three abstracts printed there; one table of contents replaces three.
5. *Write additions.* Two remarks with proofs (Remarks 10.2 and 28.2), the dated notes, the three “Further questions and research” subsections and this front matter. Everything else is delivered text.

Checks made at placement and at the write (5 October 2026). The placement dossier read the three proofs in full and found no gap. It re-derived the kernel substitutions, scalar equations, anchors, orbit closed forms and inverse recursions symbolically (58 identities), re-checked 25 rational premises of the certificates in exact fractions, confirmed each generating tree against brute force from the forbidden-relation definitions for $n \leq 9$, and computed the counts by its own dynamic programs to $n = 1500$: the residuals $(a_n/(C\mu^n n^{-3/2}) - 1 - c_1/n - c_2/n^2) n^3$, with the certified constants, stay bounded and converge (near -1062 , 21.04 , -489 and 26826 at $n = 1500$ for classes 214, 1509, 1953A and 830), as correct values of C , c_1 and c_2 require. Every equation and section number the manuscripts cite exists in Britt–Beaton v3, whose Section 4.2 says that its asymptotic expressions “are non-rigorous and are based only on numerical analysis”. At the write the OEIS entries were read again: Kotěšovec’s values (7 October 2021 for A279544, A279567 and A279569; 13 January 2026 for A279558) lie inside the certified intervals for A279544 and A279567; for A279569, and for A279558 in Kotěšovec’s normalization $c = C\sqrt{\pi}$, the displayed decimals are truncations that agree digit for digit with the certified intervals as far as they are displayed. The write also recomputed classes 1953A, 1953B and 830 by brute force for $n \leq 10$ (Remarks 10.2 and 28.2). These are checks, not a referee’s review or a formal verification. An independent adversarial check made after the write is recorded at the end of Section 30.1 (dated note, 5 October 2026).

Relation to neighbouring reports and to the repository

Two sibling reports in the same collection directory (`SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`), placed in the same commit, treat other Britt–Beaton classes by unrelated methods: `a279571-inversion-cone-walk` (A279571, bundle Reports 125 and 123: a quadrant cone walk with an irrational exponent and non- D -finiteness) and `a279551-inversion-log-deficit` (A279551 and A279556, bundle Reports

126, 127 and 129: stretched-exponential deficits, refuting Britt–Beaton’s numerical $n^{3/8}$ forms). Neither shares a lemma with this report; the three have only the source paper, the definition of inversion sequences and the existence of a threshold inverse in common. No other report treats A279544, A279567, A279569 or A279558 (searched 5 October 2026). The threshold inverses here are instances of the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` [9], as described in “The common pipeline”. Placement in the collection confers no formal status: no Lean or Rocq file in the repository mentions inversion sequences, and no statement of this report is formalized.

Part I

Inversion sequence asymptotics from iterated kernels

Bundle Report 118, archive `Inversion_Sequences_A279544_A279567_Asymptotics_and_Inverses_Source.zip`, delivered as `report118.tex` (18 pages), dated 2 October 2026. Delivered title block “REPORT 118 / Inversion sequence asymptotics from iterated kernels / Rigorous asymptotics and counting index inverses / 2 October 2026 / Self-contained proof and exact reproducibility companion”. Printed as delivered, with **[write]** notes; section, statement and equation numbers are those of the delivery. Subsection 9.1 is added in the write.

Reading conventions of Part I	Elsewhere in the report
a_n, b_n : counts of A279544, A279567; $a(z, x)$, $b(z, x)$: orbit multiplier and forcing (10)	Parts II, III: b_m, b_j are Puiseux coefficients (here $h_m, \widehat{h}_j$)
c_j relative, d_j logarithmic, e_j inverse corrections; but $d_j = v_j^{\text{orb}}$ in Section 4	Parts II, III: d_j relative, ℓ_j logarithmic; Part II: e_j, c_j orbit values
Target Y , centre $u(Y)$, $L = \log \mu$, $\beta = \frac{3}{2}$; also the bound $Y = 177/250$	Part II: $T, u(T)$, λ, p ; Part III: $M, L = \log M$
$t = \sqrt{1 - 4z}$; in Section 6 $t = \sqrt{1 - z/\sigma}$	Parts II, III: $t = \sqrt{1 - z/\rho}$, $\rho = 4/27$
$K(z, x) = 1 - x + zx + zx^2$, $M = 483/200$, radius constants R	Part II: $K = (x - 1)^2 - zx^3$, $M = 181/100$, $R(z, X)$; Part III: $R(z)$, target M
$F(z, x) = B(z, x, 0)$; $\mathcal{F}$ the class-1509 quotient; $H(t)$ a local series	Part III: $F(z)$ the generating function; Parts II, III: $H(z, x) = B(z, x, 1)$

Abstract of this Part (as delivered). For the inversion-sequence classes counted by OEIS A279544 and A279567, we prove $a_n \sim C4^n n^{-3/2}$ and $b_n \sim \widehat{C}(3 + 2\sqrt{2})^n n^{-3/2}$ with positive, explicitly characterized constants. These establish the leading laws stated numerically by Britt and Beaton for their classes 214 and 1509. Normally convergent Möbius-orbit sums from their exact catalytic equations give global continuation, convergent local square-root expansions, and exact certificates of nonzero amplitudes. The second class also requires a global nonvanishing denominator argument for an iterated quotient. Both representations give coefficient asymptotics to every fixed algebraic order, with explicit recurrences. We derive Lambert- W_{-1} counting-index inverses and shrinking two-ceiling brackets for exact integer thresholds. The local expansion is convergent; the large-index expansion is an all-fixed-order asymptotic statement. No complete exponentially improved transseries, effective numerical inverse cutoff, nonalgebraicity theorem, literature-priority claim, or external peer-review claim is made.

1 Model and main conclusions

1.1 The counted objects and source statement

An inversion sequence of length n is a vector $e = (e_1, \dots, e_n)$ of integers with $0 \leq e_i < i$. Let a_n count those vectors for which there are no indices $i < j < k$ satisfying

$$e_j \geq e_k \quad \text{and} \quad e_i \geq e_k. \quad (1)$$

Set $a_0 = 1$ and $A(z) = \sum_{n \geq 0} a_n z^n$. This is an ordinary generating function, with no factorial normalization. The sequence is OEIS A279544 [2]; its initial terms are

1, 1, 2, 4, 10, 26, 73, 214, 651, 2040, 6549, 21453, 71485, 241702, 827603.

The relation notation is $(-, \geq, \geq)$, where the dash imposes no condition on e_i, e_j . Equivalently, the forbidden reduced patterns are 000, 010, 100, 110, 120, 210.

Britt and Beaton [1, Section 3.5] give an exact auxiliary generating tree and catalytic equation for this class. Their label “214” is the length-seven count, not a shift in the sequence index. Their Section 4.2, equation (4.8), predicts $a_n \sim C4^n n^{-3/2}$ with $C \approx 0.0549097$; the section explicitly identifies its asymptotics as numerical and non-rigorous. We use version 3, dated 29 September 2026. The proof below starts from the exact enumerative model and establishes this specific leading law together with stronger fixed-order and inverse statements. Section 6 gives the same complete treatment of class 1509, whose growth constant is $3 + 2\sqrt{2}$.

[write] **5 October 2026.** Checked at placement against arXiv:2512.21943v3 (last revised 29 September 2026): its Section 4.2, “Non-algebraic classes”, says that its asymptotic expressions are non-rigorous and based only on numerical analysis, and its (4.8) and (4.10) are the class-214 and class-1509 laws quoted here, with $C \approx 0.0549097$ and $C \approx 0.0660857$. Every section and equation number this Part cites exists in v3, and the trees (4) and (34) agree with brute force from the forbidden relations for $n \leq 9$. Parts II and III prove the laws (4.11) and (4.9) of the same section by the same method (see “The common pipeline” in the front matter).

The source places the class among generating functions believed to be nonalgebraic. Neither that belief nor non- D -finiteness is proved here. Having a local square-root singularity is compatible with either an algebraic or a nonalgebraic generating function.

[write] **5 October 2026.** The belief is Britt and Beaton’s: for both classes they write that they “do not believe” the generating function to be algebraic (their Sections 3.5 and 3.6), and their Section 4.2 groups the classes it treats as those “which we believe do not have algebraic generating functions”. It is item 2 of Section 9.1.

1.2 The result and its scope

Theorem 1.1. *The generating function A has radius of convergence $\rho = 1/4$. It extends holomorphically to*

$$\{z : |z| < R\} \setminus [1/4, R) \quad (1/4 < R < 1/2), \quad (2)$$

and $1/4$ is its unique singularity on the circle of convergence. For every fixed integer $K \geq 0$,

$$a_n = C4^n n^{-3/2} \left(1 + \sum_{j=1}^K \frac{c_j}{n^j} + O(n^{-K-1}) \right), \quad n \rightarrow \infty, \quad (3)$$

where $C > 0$. The constants are uniquely specified by the convergent orbit representation and explicit recurrences below. Moreover the minimum counting index $N(Y) = \min\{n \geq 0 : a_n \geq Y\}$ admits the Lambert- W_{-1} expansion and two-ceiling brackets of Theorem 7.1.

The constants hidden in (3) may depend on K . No assertion of convergence as $K \rightarrow \infty$ is part of this theorem. The positive amplitude is certified without fitting coefficients.

The proof has four separate requirements: identify the formal generating function; prove normal convergence and global continuation; prove that the square-root term is genuinely nonzero; and transfer with the correct remainder before inverting. A kernel discriminant alone meets none of the last three requirements.

2 Exact generating tree and kernel cancellation

2.1 Auxiliary labels and coefficient extraction

The auxiliary class of [1, Section 3.5] permits the patterns 000, 010, 100 only when their zero symbols have actual value zero. Its labels (p, s) are lengths of the initial and possible second run of zeros. The root has label $(0, 0)$. Each occurrence of a label creates the following children, retaining all multiplicities:

$$\begin{aligned} (p, s) &\mapsto (p+1, s), \\ (p, s) &\mapsto (p+1, s-1) \quad (s > 0), \\ (p, 0) &\mapsto (p-\ell, \ell) \quad (0 \leq \ell \leq p-1), \\ (p, 0) &\mapsto (p-\ell-1, \ell) \quad (0 \leq \ell \leq p-2). \end{aligned} \tag{4}$$

An empty index range contributes no child. The desired objects have $s = 0$ and $p \leq 2$. This completely specifies an integer state recurrence for a_n , and is source equation (3.31).

Let $B(z, x, y)$ record length and the two label coordinates. Put $F(z, x) = B(z, x, 0)$. Every nonroot label has $p \geq 1$, so $F(z, 0) = 1$. Write $F_p(z) = [x^p]F(z, x)$. The transitions into $(1, 0)$ give

$$F_1 = z(1 + F_1 + F_2), \quad A = 1 + F_1 + F_2 = F_1/z. \tag{5}$$

The last quotient is removable at zero.

Summing (4), or clearing the denominators in source (3.33), gives the exact catalytic relation

$$0 = (y+z)(y-x) - (y-zx-zxy)(y-x)B(z, x, y) + xz(1+y)F(z, y) - z(y+2xy-x^2)F(z, x). \tag{6}$$

Each coefficient of z is a polynomial in x, y ; thus substitutions below are justified formally after denominator clearing.

2.2 The root orbit

Define

$$f_z(x) = \frac{zx}{1-zx}, \quad P(z, x) = zx^2 + (2z-1)x + z. \tag{7}$$

Substitute $y = f_z(x)$ in (6). Its B term vanishes, and elementary algebra gives

$$F(z, f_z(x)) = P(z, x)F(z, x) + Q(z, x), \quad Q(z, x) = \frac{(1+(1-z)x)(1-z-zx)}{1-zx}. \tag{8}$$

Set $F(z, x) = 1 + xG(z, x)$. Then (8) becomes

$$G(z, f_z(x)) = a(z, x)G(z, x) + b(z, x), \tag{9}$$

where

$$a(z, x) = \frac{(1-zx)P(z, x)}{z}, \quad b(z, x) = x(1-z-zx), \quad A(z) = G(z, 0)/z. \tag{10}$$

The apparent z^{-1} is addressed explicitly when its iterated arguments are substituted; it causes no unproved analytic assumption.

The small root of P is

$$r(z) = \frac{1-2z-\sqrt{1-4z}}{2z} = \frac{1-\sqrt{1-4z}}{1+\sqrt{1-4z}} \in z\mathbb{Q}[[z]]. \tag{11}$$

For an arbitrary seed x , the iterates have the exact form

$$x_j = f_z^j(x) = \frac{z^j x}{1 - zx(1 - z^j)/(1 - z)} \quad (j \geq 0). \quad (12)$$

It follows either by induction or by summing the geometric series in the denominator. Define the finite sums and their intended limit by

$$\Phi_M(z, x) = \sum_{k=0}^M b(z, x_k) \prod_{j=k+1}^M a(z, x_j), \quad (13)$$

$$\Phi(z, x) = \sum_{k \geq 0} b(z, x_k) \prod_{j > k} a(z, x_j). \quad (14)$$

An empty product is one. No reciprocal of an a factor is used. The finite sums satisfy

$$\Phi_0 = b(z, x), \quad \Phi_M = a(z, x_M) \Phi_{M-1} + b(z, x_M). \quad (15)$$

Proposition 2.1. *As an identity of formal power series,*

$$A(z) = \Phi(z, r(z))/z. \quad (16)$$

Proof. Use seed $x = r(z)$. The initial factor $a(z, r) = 0$, so repeated use of (9) gives $G(z, x_{M+1}) = \Phi_M(z, r)$. Formula (12) gives $x_j = O(z^{j+1})$. For $j \geq 1$, $a(z, x_j) = 1 + O(z^j)$, while $b(z, x_j) = O(z^{j+1})$. All products and sums therefore stabilize coefficientwise, and $G(z, x_{M+1}) \rightarrow G(z, 0)$. Now use (10). In particular, this argument never assumes that the original catalytic series $F(z, x)$ converges at the eventual critical seed $x = 1$. $\square$

3 Normal convergence and dominant singularity control

3.1 A holomorphic two-variable orbit sum

Lemma 3.1. *Choose $0 < R < 1/2$ and $0 < X < (1 - R)/R$. The series (14) and its products converge normally on $|z| \leq R$, $|x| \leq X$. They define a jointly holomorphic function near each compact set with strict room in these inequalities.*

Proof. Put $\delta = 1 - RX/(1 - R) > 0$. The orbit denominator in (12) has modulus at least δ , since $|z + \dots + z^j| \leq R/(1 - R)$. Consequently

$$|x_j| \leq (X/\delta)|z|^j.$$

For $j \geq 1$, x_j/z is holomorphic at zero and uniformly bounded by $(X/\delta)R^{j-1}$. Expand the first factor as

$$a(z, u) = 1 + (2 - z - 1/z)u + (2 - 2z)u^2 - zu^3. \quad (17)$$

There are constants K, K' independent of j, k such that

$$|a(z, x_j) - 1| \leq KR^{j-1} \quad (j \geq 1), \quad |b(z, x_k)| \leq K'R^k \quad (k \geq 0).$$

The infinite products converge normally and are bounded by $\exp(K/(1 - R))$. The outer sum is therefore dominated by a geometric series. Uniform limits on slightly larger polydiscs give joint holomorphy and termwise differentiation on the chosen compact set. Zeros in finite factors cause no problem because there is no division by such factors. $\square$

At $z = 0$, the first composed factor $a(z, x_1)$ tends to $1 - x$, rather than to one; later factors tend to one. Thus $\Phi(0, x) = x(1 - x)$. This is consistent with the lemma. Since $r(0) = 0$, the composition $\Phi(z, r(z))$ vanishes at zero and its quotient in (16) is holomorphic there.

3.2 Continuation on a slit disc

Use the principal square root on $\mathbb{C} \setminus [1/4, \infty)$. Then $\operatorname{Re} \sqrt{1 - 4z} > 0$, so the Cayley transform (11) satisfies $|r(z)| < 1$. For any $1/4 < R < 1/2$, one can choose $1 < X < (1 - R)/R$. Lemma 3.1 now proves that (16) extends the exact germ A to (2). In particular, the entire circle $|z| = 1/4$ except $z = 1/4$ is crossed holomorphically. This global argument rules out unexamined competing singularities; inspection of the kernel roots alone would not do so.

Put $t = \sqrt{1 - 4z}$ and define

$$H(t) = \frac{4}{1 - t^2} \Phi \left(\frac{1 - t^2}{4}, \frac{1 - t}{1 + t} \right) = \sum_{m \geq 0} h_m t^m. \quad (18)$$

The lemma applies near $(z, x) = (1/4, 1)$, so H is holomorphic for $|t|$ sufficiently small. This Taylor series is genuinely convergent. It follows that $A(z) = H(\sqrt{1 - 4z})$ on the local slit neighborhood, and differentiation at $t = 0$ gives

$$h_1 = -8\Phi_x(1/4, 1). \quad (19)$$

It remains to show that this coefficient is nonzero.

4 Positive amplitude by exact arithmetic

4.1 Rational critical recurrence and a proved tail

At $z = 1/4$ and seed $x = 1$, the orbit and its seed derivative are

$$u_j = \frac{3}{2 \cdot 4^j + 1}, \quad v_j^{\text{orb}} = \partial_x x_j|_{x=1} = \frac{9 \cdot 4^j}{(2 \cdot 4^j + 1)^2}. \quad (20)$$

For this subsection write $d_j = v_j^{\text{orb}}$ and

$$a(u) = (1 - u/4)(1 - u)^2, \quad b(u) = u(3 - u)/4.$$

Then

$$a'(u) = -\frac{3}{4}(1 - u)(3 - u), \quad b'(u) = (3 - 2u)/4.$$

If $q_M = \Phi_M(1/4, 1)$ and $v_M = \partial_x \Phi_M(1/4, 1)$, (15) gives the entirely rational recurrence

$$\begin{aligned} q_0 &= \frac{1}{2}, & v_0 &= \frac{1}{4}, \\ q_m &= a(u_m)q_{m-1} + b(u_m), \\ v_m &= a(u_m)v_{m-1} + d_m(a'(u_m)q_{m-1} + b'(u_m)) \quad (m \geq 1). \end{aligned} \quad (21)$$

The derivative uses the old value q_{m-1} .

For $0 \leq u \leq 1$, one has

$$0 \leq a \leq 1, \quad 0 \leq b \leq 3u/4, \quad |a'| \leq 9/4, \quad |b'| \leq 3/4, \quad 1 - a \leq 9u/4.$$

For $j \geq 1$, $u_j \leq (3/2)4^{-j}$ and $d_j \leq (9/4)4^{-j}$. In particular $q_M \leq 7/8 \leq 9/8$, and the convenient uniform bound $|a'q + b'| \leq 105/32$ yields

$$|v_M| \leq \frac{1}{4} + \frac{105}{32} \sum_{j \geq 1} d_j \leq \frac{347}{128}.$$

Therefore

$$|v_m - v_{m-1}| \leq \frac{347}{128} \frac{9}{4} u_m + \frac{105}{32} d_m \leq \frac{16929}{1024} 4^{-m}.$$

Normal convergence identifies the derivative limit, and summation proves

$$|\Phi_x(1/4, 1) - v_M| \leq E_M := \frac{16929}{3072 \cdot 4^M}. \quad (22)$$

At $M = 8$, direct exact fraction arithmetic gives

$$v_8 > 0.02433, \quad E_8 < 0.000085,$$

so $\Phi_x(1/4, 1) > 0.024245 > 0$. The finite inequalities are reproducible from (21), and no floating-point sign decision is needed. Consequently

$$C = -\frac{h_1}{2\sqrt{\pi}} = \frac{4\Phi_x(1/4, 1)}{\sqrt{\pi}} > 0. \quad (23)$$

Together with (19), this proves that $1/4$ is an actual square-root singularity. The continuation already established proves the radius and uniqueness assertions in Theorem 1.1.

4.2 Outward rational numerical enclosures

A certified decimal enclosure for (23) requires bounds on π , not just high-precision evaluation. Use Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239).$$

For either arctangent, its alternating rational series is enclosed by the sum of the first m terms and that sum plus its next term. Taking $m = 100$ and $m = 32$, respectively, gives exact rational bounds $p_- < \pi < p_+$. For $S = 10^{80}$ put

$$s_- = \frac{\lfloor \sqrt{[S^2 p_-]} \rfloor}{S}, \quad s_+ = \frac{\lfloor \sqrt{[S^2 p_+]} \rfloor + 1}{S}.$$

Integer square roots give $s_-^2 \leq p_- < \pi < p_+ < s_+^2$. Using (22) at $M = 96$, the exact interval

$$\frac{4(v_{96} - E_{96})}{s_+} < C < \frac{4(v_{96} + E_{96})}{s_-}$$

is narrower than 10^{-56} . Outward decimal rounding gives

$$\begin{aligned} 0.0549097036253448014962069269284638611865763295943683 &< C, \\ C &< 0.0549097036253448014962069269284638611865763295943684. \end{aligned} \quad (24)$$

The interval, rather than any rounded midpoint, is the certified claim. The matching high-precision numerical constant already appears in A279544, credited to Václav Kotěšovec on 7 October 2021 [2]. The contribution here is proof and certification of the amplitude, not discovery of previously unreported digits.

[write] 5 October 2026. The OEIS value was read again at the write: $c = 0.0549097036253448014962069269284638611865763295943683310517\dots$ (Kotěšovec, 7 October 2021) lies inside (24). The certified interval adds certainty, not digits.

5 Every fixed coefficient order

5.1 Transfer of the convergent local expansion

For nonintegral α , the exact binomial coefficient identity is

$$[z^n](1 - 4z)^\alpha = 4^n \frac{\Gamma(n - \alpha)}{\Gamma(-\alpha)\Gamma(n + 1)}. \quad (25)$$

The slit disc provides a standard Δ -domain at $1/4$. Singularity analysis [6, Chapter VI], applied to the convergent Taylor expansion (18), gives, for every fixed $m \geq 1$,

$$a_n = 4^n \sum_{j=0}^{m-1} \frac{h_{2j+1}}{\Gamma(-j - 1/2)} \frac{\Gamma(n - j - 1/2)}{\Gamma(n + 1)} + O(4^n n^{-m-3/2}). \quad (26)$$

For clarity about the remainder, expand H through degree $2m + 2$. The even terms are polynomials in z and have zero coefficients for all sufficiently large n . The first omitted odd term contributes $O(4^n n^{-m-3/2})$; the remaining Taylor remainder has still smaller transfer order. No differentiation of an uncontrolled asymptotic remainder is involved.

Define the standard gamma-ratio polynomials by

$$\frac{\Gamma(n - \alpha)}{\Gamma(n + 1)} \sim n^{-\alpha-1} \sum_{j \geq 0} g_j(\alpha) n^{-j}, \quad g_0(\alpha) = 1. \quad (27)$$

Then the exact prescription for the coefficients in (3) is

$$C_m := C c_m = \sum_{k=0}^m \frac{h_{2k+1} g_{m-k}(k + 1/2)}{\Gamma(-k - 1/2)}, \quad c_0 = 1. \quad (28)$$

In particular, $C_0 = C$. For a fully explicit recurrence, let $B_j(x)$ be the Bernoulli polynomials, defined by $w e^{xw} / (e^w - 1) = \sum_{j \geq 0} B_j(x) w^j / j!$. Set

$$\lambda_j(\alpha) = \frac{(-1)^{j+1} (B_{j+1}(-\alpha) - B_{j+1}(1))}{j(j+1)}, \quad g_m(\alpha) = \frac{1}{m} \sum_{j=1}^m j \lambda_j(\alpha) g_{m-j}(\alpha). \quad (29)$$

This follows by exponentiating the logarithmic Stirling expansion of the ratio in (27). It determines any fixed order using rational arithmetic at the half-integer arguments.

The first two relative corrections are

$$\begin{aligned} c_1 &= \frac{3}{8} - \frac{3}{2} \frac{h_3}{h_1}, \\ c_2 &= \frac{25}{128} - \frac{45}{16} \frac{h_3}{h_1} + \frac{15}{4} \frac{h_5}{h_1}. \end{aligned} \quad (30)$$

Here $g_1(1/2) = 3/8$, $g_1(3/2) = 15/8$, and $g_2(1/2) = 25/128$. These identities also provide immediate normalization checks.

5.2 Certified local jets and the first corrections

The Taylor coefficients h_j can be computed by rational series arithmetic in (15), after substituting $z = (1 - t^2)/4$ and $x = (1 - t)/(1 + t)$. A uniform complex-circle bound certifies the truncation. On $|t| = 1/10$ choose

$$R = 101/400, \quad X = 11/9, \quad \delta = 1 - RX/(1 - R), \quad T = X/\delta.$$

Then $|z| \leq R$, $|x| \leq X$ and $|x_j| \leq TR^j$. Define

$$\begin{aligned} A_* &= T + (2 + R)TR + (2 + 2R)T^2R^2 + T^3R^3, \\ B_* &= T(1 + R + RT). \end{aligned}$$

Equation (17) gives $|a(z, x_j) - 1| \leq A_*R^{j-1}$ for $j \geq 1$, and $|b(z, x_j)| \leq B_*R^j$. Exact rational inequalities give

$$A_*/(1 - R) < 6, \quad B_* < 4, \quad (1 - R)^{-1} < 3/2.$$

Using $e < 3$, the finite sums obey $|\Phi_M| \leq B_*e^{A_*/(1-R)}/(1 - R) < 5000$. Telescoping (15) therefore yields

$$|\Phi - \Phi_M| \leq \frac{5000A_* + B_*R}{1 - R}R^M < 40000R^M.$$

Since $|1/z| \leq 400/99 < 5$, if $H_M(t) = \Phi_M(z(t), x(t))/z(t)$, then

$$\sup_{|t|=1/10} |H - H_M| < 200000R^M, \quad |h_j - [t^j]H_M| < 200000 \cdot 10^j R^M. \quad (31)$$

The second inequality is Cauchy's estimate. This supplies an effective certificate for every fixed local coefficient by increasing M .

Degree-five rational jets with $M = 64$, followed by interval arithmetic in (30), give the outward enclosures

$$\begin{aligned} 27.604095556555475840191975 &< c_1 < 27.604095556555475840191976, \\ 43.988908237439005751766276 &< c_2 < 43.988908237439005751766277. \end{aligned} \quad (32)$$

For orientation, $h_1 \approx -0.19464983128564593535$, $h_3 \approx 3.53342590409616985565$, and $h_5 \approx 0.37689182228965126927$. The comparatively large c_1 warns against interpreting a short moderate- n fit as a proof of the leading constant.

5.3 What is and is not an all-order result

The series (18) converges in a neighborhood of $t = 0$. The expansion (3) means that each fixed truncation has its own proved algebraic remainder. These are different statements. No uniform estimate in a truncation order growing with n has been proved. We do not claim optimal truncation, all subdominant exponential sectors, Stokes constants, or a complete resurgent transseries. Nor do the finite coefficient checks replace the analytic continuation and tail arguments above.

6 A quotient kernel for class 1509

6.1 The second counted class and exact equations

Let b_n count inversion sequences with no $i < j < k$ satisfying

$$e_j \geq e_k \quad \text{and} \quad e_i > e_k. \quad (33)$$

This is class $(-, \geq, >)$, or avoidance of 100, 110, 120, 210, counted by OEIS A279567 [3]. Put $\mathcal{A}(z) = \sum_{n \geq 0} b_n z^n$. Its initial terms are

$$1, 1, 2, 6, 21, 82, 343, 1509, 6893, 32419, 156058, 765578, 3815062.$$

Source [1, Section 3.6, equation (3.34)] gives the root $(0, 0)$ and these multiplicity-preserving transitions:

$$\begin{aligned} (p, s) &\mapsto (p+1, s), \\ (p, s) &\mapsto (p+1, s-1) \quad (s > 0), \\ (p, s) &\mapsto (p-i, 0) \quad (s \leq 1, 0 \leq i < p), \\ (p, s) &\mapsto (p-\ell, \ell-k) \quad (s \leq 1, 1 \leq \ell < p, 0 \leq k < \ell). \end{aligned} \quad (34)$$

The target is the sum of all states with $s \leq 1$. The auxiliary class allows 100 when both zero symbols have actual value zero. Let $\mathcal{B}(z, x, y)$ be its generating function, and set

$$J(z, x) = \mathcal{B}(z, x, 0) + [y]\mathcal{B}(z, x, y), \quad \mathcal{A}(z) = J(z, 1).$$

Write $B_0(x) = \mathcal{B}(z, x, 0)$ and $B_1(x) = [y]\mathcal{B}(z, x, y)$, suppressing only the z argument. The source's equations (3.36)–(3.37) are, equivalently,

$$B_0(x) = 1 + zxJ(z, x) + \frac{zx}{1-x}(\mathcal{A} - J(z, x)), \quad (35)$$

$$\begin{aligned} \mathcal{B}(z, x, y) &= B_0(x) - zx B_1(x) + \frac{zx(y+1)}{y}(\mathcal{B}(z, x, y) - B_0(x)) \\ &\quad - \frac{zxy}{(1-x)(x-y)}J(z, x) + \frac{zxy}{(1-x)(1-y)}\mathcal{A} + \frac{zxy}{(1-y)(x-y)}J(z, y). \end{aligned} \quad (36)$$

Clear denominators before making formal substitutions. Equation (35) gives $B_0(x) = 1 + zx\mathcal{A}/(1-x) - zx^2J(z, x)/(1-x)$. Set $y = f_z(x) = zx/(1-zx)$ in (36), substitute this expression, and collect terms. The result is

$$J(z, f_z(x)) = c(z, x)J(z, x) + d(z, x) + e(z, x)\mathcal{A}(z), \quad (37)$$

where, with $K(z, x) = 1 - x + zx + zx^2$,

$$\begin{aligned} c(z, x) &= \frac{(2zx-1)K(z, x)}{z(x-1)}, \\ d(z, x) &= \frac{(2zx-1)(zx+z-1)}{z(zx-1)}, \quad e(z, x) = -\frac{2x(zx+z-1)}{x-1}. \end{aligned} \quad (38)$$

Thus the scalar reduction follows by rational algebra from the exact source model, not from a guessed coefficient recurrence.

Define

$$\sigma = 3 - 2\sqrt{2}, \quad \mu = \sigma^{-1} = 3 + 2\sqrt{2}, \quad X(z) = \frac{1 - z - \sqrt{1 - 6z + z^2}}{2z}, \quad X(0) = 1. \quad (39)$$

It satisfies $X = 1 + zX + zX^2$, so its Taylor coefficients are nonnegative. Put

$$y(z) = \frac{X(z) - 1}{2} = f_z(X(z)), \quad X(\sigma) = 1 + \sqrt{2}, \quad y(\sigma) = 1/\sqrt{2}. \quad (40)$$

At $x = X(z)$, the kernel K vanishes. The substitution is legitimate because every coefficient in z of J is a polynomial in x ; intermediate divisions by $X - 1 = 2z + O(z^2)$ are interpreted in Laurent series, or removed by clearing denominators. At that root, $d = y - 1/y$ and $e = 1/y$, whence

$$\mathcal{A} = 1 - y^2 + yJ(z, y). \quad (41)$$

6.2 Iterating in the contracting direction

Rearrange (37) in the opposite direction from class 214:

$$J(z, x) = p(z, x)J(z, f_z(x)) + q(z, x) + r_*(z, x)\mathcal{A}(z), \quad (42)$$

where

$$\begin{aligned} p(z, x) &= \frac{z(1 - x)}{(1 - 2zx)K(z, x)}, \\ q(z, x) &= \frac{(1 - x)(1 - z - zx)}{(1 - zx)K(z, x)}, \\ r_*(z, x) &= \frac{2zx(1 - z - zx)}{(1 - 2zx)K(z, x)}. \end{aligned} \quad (43)$$

For the orbit $x_j = f_z^j(x)$ of (12), put $P_0 = 1$, $P_j = \prod_{i=0}^{j-1} p(z, x_i)$, and define

$$U(z, x) = \sum_{j \geq 0} P_j q(z, x_j), \quad V(z, x) = \sum_{j \geq 0} P_j r_*(z, x_j). \quad (44)$$

Let U_N, V_N stop at $j = N - 1$. Finite iteration gives exactly

$$J(z, x) = U_N(z, x) + V_N(z, x)\mathcal{A}(z) + P_N J(z, x_N).$$

For $x = y(z) = O(z)$, every $p(z, x_j)$ has positive z valuation, so the last term tends coefficientwise to zero. Combining with (41) proves the formal identity

$$\mathcal{A}(z) = \mathcal{F}(z, y(z)), \quad \mathcal{F}(z, x) = \frac{1 - x^2 + xU(z, x)}{1 - xV(z, x)}. \quad (45)$$

Its denominator has constant term one. The essential additional analytic issue is to exclude zeros of this new denominator throughout the whole disc, not merely at the positive critical point.

6.3 A global nonzero denominator certificate

For $|z| \leq \sigma$, positivity and (39)–(40) give

$$|z| < R := 43/250, \quad |X(z)| < M := 483/200, \quad |y(z)| < Y := 177/250.$$

The algebraic kernel factors as

$$K(z, x) = (1 - x/X(z))(1 - zX(z)x).$$

Since X is never zero and $y/X = z/(1 - zX)$, at the first orbit point $x_0 = y(z)$ the lower bound is

$$|K(z, x_0)| \geq k_0 := (1 - a)(1 - RMY) > 0, \quad a = R/(1 - RM). \quad (46)$$

The exact rational upper bounds

$$\begin{aligned} \frac{R(1+Y)}{(1-2RY)k_0} &< \frac{4}{5}, \quad \frac{2RY(1+R+RY)}{(1-2RY)k_0} < \frac{17}{20}, \\ \frac{RY}{1-RY} &< \frac{7}{50} \end{aligned} \quad (47)$$

imply respectively $|p(z, x_0)| < 4/5$, $|r_*(z, x_0)| < 17/20$, and $|x_1| < 7/50$. Also $|q(z, x_0)| \leq (1 + Y)(1 + R + RY)/((1 - RY)k_0)$ is finite.

For $|x| \leq w := 7/50$, the simpler lower bound $|K(z, x)| \geq k_s := 1 - w - Rw - Rw^2 > 0$ suffices. Direct rational inequalities give

$$|p| < \frac{1}{4}, \quad |q| < 2, \quad |r_*| < \frac{3}{5}|x|, \quad |f_z(x)| < \frac{9}{50}|x|. \quad (48)$$

For example, the upper bounds for p and r_*/x are $R(1+w)/((1-2Rw)k_s)$ and $2R(1+R+Rw)/((1-2Rw)k_s)$; the corresponding bounds for q and $f_z(x)/x$ are $(1+w)(1+R+Rw)/((1-Rw)k_s)$ and $R/(1-Rw)$. This displays every finite inequality needed for (48).

The orbit satisfies $|x_j| \leq w(9/50)^{j-1}$ for $j \geq 1$. Normal convergence of both series (44) follows by geometric majorants. In particular,

$$|yV(z, y)| \leq Y \left(\frac{17}{20} + \frac{(4/5)(3/5)(7/50)}{1 - (1/4)(9/50)} \right) < \frac{2}{3}. \quad (49)$$

Hence $|1 - yV(z, y)| > 1/3$ throughout the closed disc. The factorization (46) is important: using only the crude first-step triangle bound would lose the strength required for this global conclusion. The formal identity (45) is therefore a convergent analytic representation in $|z| < \sigma$ without a presupposed growth estimate for the coefficients.

There is also a two-variable neighborhood for U, V . On $|z| \leq 43/250$, $|x| \leq 709/1000$, one has

$$|K(z, x)| \geq 1 - |x| - R|x| - R|x|^2 > 0, \quad |p(z, x)| < 5, \quad |f_z(x)| < 7/50.$$

The later iterates obey (48). All these inequalities have strict margins, so slightly larger polydiscs still work. Thus U, V are holomorphic on a neighborhood of this closed polydisc; $\mathcal{F}$ is holomorphic near the compact critical curve $\{(z, y(z)) : |z| \leq \sigma\}$ by (49).

Away from $z = \sigma$ on the boundary circle, the discriminant in (39) is nonzero, so X, y extend holomorphically. The nonzero denominators and contraction estimates persist locally. At σ , use

a square-root coordinate for y and the just-proved two-variable holomorphy. Compactness then supplies $\eta > 0$ and $0 < \phi < \pi/2$ such that $\mathcal{A}$ continues to the Δ -domain

$$\{z : |z| < \sigma + \eta, z \neq \sigma, |\arg(z - \sigma)| > \phi\}. \quad (50)$$

The next step proves that the possible singularity at σ is real.

[write] **5 October 2026.** Part II excludes competing singularities for class 1953A by a different argument, which Report 119 contrasts with this one in Section 10.3: a determinant gap only on the positive segment, Pringsheim's theorem and removal of boundary poles. Part III uses a third device for class 830, a zero-free orbit product on a disc larger than the circle of convergence (Proposition 24.1). The uniform majorant (49) is the closest of the three to it.

6.4 An exact positive critical derivative

Put $y_* = 1/\sqrt{2}$. For fixed $z = \sigma$, the disc $|x - y_*| \leq 1/1000$ lies in $|x| < 709/1000$. Its first product factor has modulus less than five; subsequent factors have modulus less than $1/4$. Since both subsequent q and r_* have modulus less than two, the omitted tails in (44) satisfy, for $N \geq 1$,

$$|U - U_N|, |V - V_N| \leq E_N := \frac{40}{3} 4^{1-N}. \quad (51)$$

Indeed, after the first factor the remaining geometric majorant is $2/(1 - 1/4) = 8/3$. Cauchy's formula gives at $x = y_*$

$$|U_x - (U_N)_x|, |V_x - (V_N)_x| \leq 1000 E_N. \quad (52)$$

The finite quantities are reproducible with rational interval arithmetic. Start with zero terminal values and use, backwards for $j = N - 1, \dots, 0$,

$$U_j = q(z, x_j) + p(z, x_j)U_{j+1}, \quad V_j = r_*(z, x_j) + p(z, x_j)V_{j+1}.$$

For derivatives, set $\dot{x}_0 = 1$, $\dot{x}_{j+1} = z\dot{x}_j/(1 - zx_j)^2$, and use, for example,

$$\dot{U}_j = q_x(z, x_j)\dot{x}_j + p_x(z, x_j)\dot{x}_jU_{j+1} + p(z, x_j)\dot{U}_{j+1}.$$

Use the identical rule with q, U replaced by r_*, V . All functions being differentiated are the displayed rational functions. Enclose $\sqrt{2}$ by scaled integer square roots, then form the intervals for σ and y_* . At $N = 24$, interval operations on the rational grid $10^{-60}\mathbb{Z}$, rounding every operation outward, followed by (51)–(52) and quotient differentiation in (45), give

$$0.1631959191663560 < \mathcal{F}_x(\sigma, y_*) < 0.1631959201118698. \quad (53)$$

In particular, the derivative is strictly positive. This is an exact finite interval certificate plus a proved infinite tail, not a floating-point sign test.

Set $t = \sqrt{1 - z/\sigma}$ and $z = \sigma(1 - t^2)$. The local branch is

$$y(z) = \frac{1 - 3z - t\sqrt{1 - \sigma z}}{4z} = y_* - \kappa t + O(t^2), \quad \kappa = \frac{\sqrt{1 - \sigma^2}}{4\sigma} > 0. \quad (54)$$

Since $\mathcal{F}$ is holomorphic near (σ, y_*) , the expansion

$$\mathcal{F}\left(\sigma(1 - t^2), \frac{1 - 3\sigma(1 - t^2) - t\sqrt{1 - \sigma^2(1 - t^2)}}{4\sigma(1 - t^2)}\right) = \sum_{j \geq 0} \hat{h}_j t^j \quad (55)$$

converges for sufficiently small $|t|$. In particular,

$$\hat{h}_1 = -\kappa \mathcal{F}_x(\sigma, y_*) < 0, \quad \hat{C} = \frac{\kappa \mathcal{F}_x(\sigma, y_*)}{2\sqrt{\pi}} > 0. \quad (56)$$

A rational Machin-series enclosure for π , combined with the same outward interval arithmetic, proves

$$0.0660857086342079 < \hat{C} < 0.0660857090170910. \quad (57)$$

The more precise numerical value $0.066085708825649431 \dots$ was already reported by Václav Kotěšovec in A279567 on 7 October 2021 [3]. The short certificate (57) suffices for positivity; the higher-jet certificate below proves a tighter enclosure.

[write] 5 October 2026. The OEIS value $0.066085708825649431003670013119332303648755519420440375 \dots$ (Kotěšovec, 7 October 2021), read again at the write, lies inside (57) and inside the tighter (63).

6.5 The second all-order theorem

Theorem 6.1. *For A279567, the radius of convergence is $\sigma = 3 - 2\sqrt{2}$, with σ the unique dominant singularity. For every fixed $K \geq 0$,*

$$b_n = \hat{C} \mu^n n^{-3/2} \left(1 + \sum_{j=1}^K \frac{\hat{c}_j}{n^j} + O(n^{-K-1}) \right), \quad \mu = 3 + 2\sqrt{2}. \quad (58)$$

The amplitude is (56), certified by (57). Using g_j from (29), all corrections are given by

$$\hat{C} \hat{c}_m = \sum_{k=0}^m \frac{\hat{h}_{2k+1} g_{m-k}(k+1/2)}{\Gamma(-k-1/2)}. \quad (59)$$

In particular,

$$\hat{c}_1 = \frac{3}{8} - \frac{3\hat{h}_3}{2\hat{h}_1}, \quad \hat{c}_2 = \frac{25}{128} - \frac{45\hat{h}_3}{16\hat{h}_1} + \frac{15\hat{h}_5}{4\hat{h}_1}.$$

The counting-index inverse of Section 7 applies with $4, C, c_j$ replaced by $\mu, \hat{C}, \hat{c}_j$.

Proof. The nonzero coefficient (56) makes σ a genuine square-root singularity. The convergence and continuation leading to (50) prove radius and uniqueness. Apply exactly the transfer argument of (26), replacing $4^n, h_j$ by $\mu^n, \hat{h}_j$. More explicitly, expansion through degree $2K+4$ has remainder $O(t^{2K+5})$; even powers are polynomials in z , the last retained odd term contributes at most $O(\mu^n n^{-K-5/2})$, and the remaining terms are smaller after transfer. The gamma-ratio recurrence proves (59) and (58). This also proves source equation (4.10), stated there as a numerical leading law. The convergent local series permits certification of each fixed coefficient by finite orbit jets and Cauchy bounds; no numerical evaluation of a correction is required for the theorem. $\square$

6.6 Certified correction jets for class 1509

The preceding $N = 24$ certificate is enough to prove positivity. A separate degree-five jet calculation gives a much tighter amplitude and certifies the first two relative corrections. Let $h = 1/10000$ and retain the exact local parametrization (54). On $|t| \leq h$, subtraction at zero gives

$$y(t) - y_* = \frac{t^2 - t\sqrt{1 - \sigma^2 + \sigma^2 t^2}}{4\sigma(1 - t^2)}.$$

The radicand has modulus less than one, so

$$|y(t) - y_*| \leq \frac{h^2 + h}{4\sigma(1 - h^2)} < \frac{1}{5000}. \quad (60)$$

Using the rational bounds $\sigma < 171573/10^6$, $y_* < 707107/10^6$, and $\sigma > 171/1000$, one obtains $|z| < R$, $|y| < Y$, $|X| = |1 + 2y| < M$ with the same R, M, Y used above. The identities $K(z, X) = 0$ and $y = f_z(X)$ still hold throughout this t disc, including both signs of t . Therefore the kernel factorization, first-step bound $|p_0| < 4/5$, later bound $|p_j| < 1/4$, and $|yV| < 2/3$ apply on both branches. This uses direct complex bounds, not positivity improperly extended to the second branch.

For N terms the uniform tails now satisfy

$$|U - U_N|, |V - V_N| \leq \tilde{E}_N := \frac{32}{15} 4^{1-N}.$$

The first q bound plus $(4/5)(8/3)$ is less than eight, so $|U|, |U_N| < 8$. For the numerator and denominator of (45), write $B = 1 - y^2 + yU$ and $D = 1 - yV$, and define B_N, D_N with the finite sums. Then $|B| < 8$, $|D| > 1/3$, and $|B - B_N|, |D - D_N| \leq Y\tilde{E}_N$. At $N = 96$, $Y\tilde{E}_N < 1/12$, hence $|D_N| > 1/4$. Thus

$$\left| \frac{B}{D} - \frac{B_N}{D_N} \right| \leq \frac{|B - B_N|}{|D_N|} + \frac{|B| |D - D_N|}{|D| |D_N|} < 100\tilde{E}_N. \quad (61)$$

By Cauchy's coefficient estimate,

$$\left| \hat{h}_j - [t^j] \mathcal{F}_N(z(t), y(t)) \right| \leq 100\tilde{E}_N h^{-j}. \quad (62)$$

There is no factorial in this coefficient bound.

The finite Taylor jets are evaluated with the same outward rational interval arithmetic as before. Addition and multiplication are componentwise sum and truncated convolution. If $A(t)B(t) = 1$, the reciprocal recurrence is

$$B_0 = A_0^{-1}, \quad B_n = -A_0^{-1} \sum_{k=1}^n A_k B_{n-k}.$$

For the radical in (54), its degree-four jet is

$$s + \frac{\sigma^2}{2s} t^2 - \frac{\sigma^4}{8s^3} t^4, \quad s = \sqrt{1 - \sigma^2}.$$

Multiplication by t means this is sufficient through degree five. After widening the finite jet coefficients by (62), the formulas of (56) and Theorem 6.1 give the following conservative strict enclosures:

$$\begin{aligned} 0.066085708825649431003670013119332 &< \hat{C}, \\ \hat{C} &< 0.066085708825649431003670013119333, \\ 8.0552783200743063565266608644113 &< \hat{c}_1, \\ \hat{c}_1 &< 8.0552783200743063565266608644114, \\ -17.49290139551774359177529920663 &< \hat{c}_2, \\ \hat{c}_2 &< -17.49290139551774359177529920662. \end{aligned} \quad (63)$$

These bounds certify the two corrections, as well as the indicated amplitude digits. All higher correction coefficients remain specified exactly by (59); this report numerically certifies only the first two relative corrections for each class.

7 The counting index inverse

7.1 Leading inverse and formal correction recurrence

The inverse considered here takes a target count to its required index. It is not the compositional inverse of the generating function $A(z)$. The following notation applies to either sequence: let μ_0 be its growth constant, C_0 its positive amplitude, and $c_j^{(0)}$ its relative corrections. Thus $(\mu_0, C_0, c_j^{(0)})$ equals $(4, C, c_j)$ for A279544 and $(3 + 2\sqrt{2}, \hat{C}, \hat{c}_j)$ for A279567. In this section abbreviate them to μ, C, c_j , and write $L = \log \mu$ and $\beta = 3/2$. For sufficiently large real $Y > 0$, let

$$u(Y) = -\frac{\beta}{L} W_{-1} \left(-\frac{L}{\beta} \left(\frac{C}{Y} \right)^{1/\beta} \right). \quad (64)$$

The branch W_{-1} is the real branch with values at most -1 on $[-1/e, 0)$ [7]. It selects the large solution of

$$C e^{Lu} u^{-\beta} = Y. \quad (65)$$

The principal branch would select the small solution and would be incorrect for this counting inverse. In logarithmic form,

$$u(Y) = \frac{\log Y + \beta \log \log Y - \log C - \beta \log L}{L} + O \left(\frac{\log \log Y}{\log Y} \right). \quad (66)$$

Let the logarithmic correction coefficients be defined formally by

$$\log \left(1 + \sum_{j \geq 1} c_j w^j \right) = \sum_{j \geq 1} d_j w^j. \quad (67)$$

Thus $d_1 = c_1$, $d_2 = c_2 - c_1^2/2$, and in general

$$d_m = c_m - \frac{1}{m} \sum_{j=1}^{m-1} j d_j c_{m-j}. \quad (68)$$

There is a unique formal inverse expansion

$$y(Y) = u(Y) + \sum_{j \geq 1} e_j u(Y)^{-j}, \quad (69)$$

obtained by setting all powers of u^{-1} to zero in

$$L(y - u) - \beta \log(y/u) + \sum_{j \geq 1} d_j y^{-j}. \quad (70)$$

At order u^{-m} , the new unknown occurs only as Le_m ; hence this is an unambiguous recursion. In particular,

$$\begin{aligned} e_1 &= -d_1/L, \\ e_2 &= -\beta d_1/L^2 - d_2/L, \\ e_3 &= -\beta^2 d_1/L^3 - \beta d_2/L^2 - d_1^2/L^2 - d_3/L. \end{aligned} \quad (71)$$

This is the smooth inverse of the asymptotic model. It does not impose a canonical interpolation on the discrete sequence a_n .

7.2 Strict increase and integer threshold brackets

Appending the value n to any admissible length- n sequence gives an admissible length- $(n+1)$ sequence: a new forbidden triple would have its last, maximal entry at $k = n+1$, which is impossible in (1). The sequence $(0, 1, \dots, n-1, n-1)$ is also admissible and is not in the image. For $n = 1$ it is $(0, 0)$. Therefore $a_{n+1} > a_n$ for $n \geq 1$. For class 1509, appending the maximum gives the same injection, and the all-zero sequence supplies an extra valid object. Thus $b_{n+1} > b_n$ for $n \geq 1$ as well. In the theorem below, a_n denotes either of these two sequences with its matching constants.

Theorem 7.1. *Let $N(Y) = \min\{n \geq 0 : a_n \geq Y\}$ and, for a fixed integer $K \geq 0$, put*

$$y_K(Y) = u(Y) + \sum_{j=1}^K e_j u(Y)^{-j}.$$

There are constants $E_K > 0$ and Y_K such that, for every $Y \geq Y_K$,

$$\left\lceil y_K(Y) - E_K u(Y)^{-K-1} \right\rceil \leq N(Y) \leq \left\lceil y_K(Y) + E_K u(Y)^{-K-1} \right\rceil. \quad (72)$$

For sufficiently large Y , these bounds contain at most two adjacent integer candidates. The convention at equality is $N(a_n) = n$ for $n \geq 2$.

Proof. The coefficient expansion implies, for every fixed K ,

$$\log a_n = Ln - \beta \log n + \log C + \sum_{j=1}^K d_j n^{-j} + O(n^{-K-1}).$$

Let $F_K(x) = Lx - \beta \log x + \log C + \sum_{j=1}^K d_j x^{-j}$. Choose D and a starting index so that the error at every subsequent integer is bounded in absolute value by Dx^{-K-1} . The smooth functions $F_K(x) \pm Dx^{-K-1}$ are eventually strictly increasing, because their derivatives tend to $L > 0$.

Invert these two envelopes at $\log Y$. Formal substitution from (70) gives $F_K(y_K) - \log Y = O(u^{-K-1})$; the mean value theorem, using a fixed positive lower derivative bound, puts both inverse roots at $y_K + O(u^{-K-1})$. If an integer lies below the inverse root of the upper envelope, its count is below Y . Every integer at or above the inverse root of the lower envelope has count at least Y . Applying the corresponding ceilings gives (72), after enlarging the fixed constant E_K . Since the real interval has length $2E_K u^{-K-1} \rightarrow 0$, its ceilings are eventually equal or adjacent. $\square$

Already at $K = 0$, the real bracket width is $O(1/u)$, not $O(1)$. It is essential to retain both ceilings: a target may lie arbitrarily close to an exact sequence value, so unconditional rounding of y_K or an unconditional single ceiling is not justified. The proof gives existence of E_K, Y_K , but this report supplies no effective numerical values for them. The theorem is therefore not an executable certified threshold routine for a specified finite target.

[write] **5 October 2026.** *The same theorem in the other Parts.* Part II, Theorem 17.1, is this theorem for A279569, with

$$(\mu, C, c_j, d_j, e_j, Y, y_K, E_K, Y_K, L, \beta) \quad \text{read as} \quad (27/4, C, d_j, \ell_j, \eta_j, T, v_K, M_K, T_K, \lambda, p);$$

its η_1, η_2 (126) are (71) under this dictionary, and its proof is a second write-up of the argument above. Part III, Proposition 28.1, is the same bracket for A279558 in the coordinate $L = \log M$, with the Lambert centre expanded in powers of $1/L$ (see the note after that proposition).

Repository context. The theorem is an instance of results of the transseries volume [9], which it neither cites nor needs. With $a = L = \log \mu$ and $b = -\beta$, equation (65) reads $aX + b \log X = \log Y - \log C$; the branch rule of `p0:thm:lambert-core` for $b < 0$ gives its large root as $(b/a)W_{-1}(-(a/|b|)e^{-(\log Y - \log C)/|b|})$, which is (64). The expansion (69), determined by (70), has the form of the all-orders reversion around the Lambert core, `p0:thm:lambert-centered`. The counting index $N(Y)$ is, for $Y > 1$, the integer staircase of `p0:def:three-inverses` with $n_1 = 1$, since $a_0 = a_1 = 1$ and the counts increase strictly from $n = 1$ (precision added 5 October 2026); a bracket whose two ceilings agree determines it, as in the separation condition of `p0:thm:staircase`; and the mean-value step of the proof is the estimate of `p0:cor:forward-to-inverse`, applied to each envelope $F_K \pm Dx^{-K-1}$ instead of to an admissible interpolation of a_n . These are pointers to known mechanics, not a second proof of the volume’s theorems.

8 Reproducibility and limitations

The companion package contains this standalone L^AT_EX source, the rendered PDF, portable exact checks and fixed data, and deterministic build and archive scripts. The checks distinguish the 29 displayed OEIS A279544 prefix terms and 26 displayed A279567 prefix terms from longer internally generated coefficient lists. Direct brute-force counting through $n = 8$ independently verifies both forbidden-relation models. For each class, an exact orbit-series construction and a state recurrence independently agree at all 81 indices $n = 0, \dots, 80$. The class-1509 reconstruction uses the rational orbit quotient. These finite checks identify the models and catch algebraic or indexing mistakes; they do not prove the asymptotic assertions by sampling.

The amplitude check uses rational recurrence (21), its proved infinite tail (22), rational arctangent intervals, and exact integer square roots. The class-214 correction check uses rational jets and the uniform analytic estimate (31). The class-1509 checks certify the global denominator bound (49), the critical derivative with its Cauchy tail, and the higher-jet bounds (61)–(62). Every decimal interval is rounded outward and checked against its rational endpoints. The mathematical justification for infinite convergence, continuation and coefficient transfer is in this report.

The README describes the precise commands, tested toolchain, fixture schema, checks and failure tests. The package inventory is closed: changed, missing and unexpected files are rejected. Checksums detect changes but are not authenticated signatures. The reproducibility claim is same-toolchain replay in fresh directories, rather than identity across arbitrary TeX, font or compression versions.

The result establishes the source’s class-214 and class-1509 leading asymptotic laws and all fixed algebraic orders, with their counting-index inverses in the stated sense. It does not establish nonalgebraicity or non- D -finiteness, solve every neighboring inversion class, or supply a complete exponentially improved expansion. The source comparison is explicit and bounded; no global literature-priority or external peer-review claim is made.

[write] 5 October 2026. In this repository the companion package is shipped under the prefix `118-kernels-` in `code/` and `data/`, with `118-kernels-checks-README.md`; the PDF, `CHECKSUMS.sha256` and `checks/MANIFEST.json` are not shipped, and “the README” above is the delivered `README.txt`, which this report’s README replaces. Because the inventory is closed and the scripts read the delivered layout, the checks are rerun from the archive, not in this directory; the report’s README gives the route and records which runs passed.

9 Further research questions

The proofs isolate several concrete next questions without presuming their answers.

1. **Effective inverse thresholds.** Can one turn the coefficient-transfer remainder into explicit constants E_K, Y_K in (72)? This would give a finite-target certified inverse routine, with exact recurrence checks only at the remaining one or two candidate indices.
2. **Algebraic and differential status.** Do these generating functions fail to be algebraic or D -finite, as suggested in the source? The present local square-root descriptions do not decide either question.
3. **Secondary singularities and exponential improvement.** Can the orbit representations be continued farther, and can their secondary singularities and exponentially small coefficient contributions be classified? The bound $R < 1/2$ in the class-214 continuation proof is a sufficient domain for this argument, not a claim of a singularity at $1/2$.
4. **Other catalytic kernels.** Which additional inversion-sequence classes admit a contracting orbit representation with an equally effective nonzero-denominator certificate? The class-1509 quotient illustrates a new obstruction that must be handled before transferring a kernel's square root.

9.1 Further questions and research

[write] (5 October 2026.) Under Vladimir's standing rule of 4 October 2026, this subsection collects every claim of Part I that is stated without proof or left open, with its source, the sketch offered and what is missing. Nothing in this Part was found to be wrong, and no claim was refuted.

1. *Effective inverse thresholds* (question 1; the paragraph after Theorem 7.1: E_K, Y_K are existential). Sketch offered: make the coefficient-transfer remainder explicit. Missing: an explicit Δ -domain with bounds on the local expansion there, explicit transfer constants, and from them explicit envelopes; the finite search at the one or two candidate indices would then be exact. Parts II and III ask the same for their classes.
2. *Algebraic and D -finite status* (question 2; Section 1.1). Source: Britt and Beaton believe both generating functions nonalgebraic. The square-root singularity is compatible with either answer, and the orbit representations are infinite. Missing: any global argument (a natural boundary, an infinite set of singularities, or an algebraic or differential equation).
3. *Secondary singularities, exponential improvement and maximal continuation* (question 3; Section 5.3). Sketch: continue the orbit representations further; $R < 1/2$ is only a sufficient domain. Missing: the location and type of further singularities, the exponentially small contributions they give, and the non-claims of Section 5.3: convergence of the $1/n$ series as the order grows, optimal truncation, Stokes constants and a resurgent transseries.
4. *Higher corrections.* The coefficients c_j and $\hat{c}_j$ for $j \geq 3$ are specified exactly by (28) and (59) and computable to any accuracy by (31) and (62), but only the first two of each are certified numerically (Section 6, last paragraph). Their sign pattern and growth are open; Part II asks the same (its item 3).

5. *Other catalytic kernels* (question 4). Parts II and III give two further instances with different continuation devices (a determinant gap; a zero-free product); a criterion for which classes admit a contracting orbit with an effective nonvanishing certificate is open.
6. *Earlier rigorous proofs*. This Part claims no literature priority. The placement and the write checked the Britt–Beaton and OEIS statements cited, but did not search the wider literature for an earlier rigorous proof of these laws; that bibliographic question is open.

Part II

Inversion sequence asymptotics from two kernel roots

Bundle Report 119, archive `A279569_Two_Kernel_Asymptotics_and_Inverses_Source.zip`, delivered as `report119.tex` (16 pages), dated 2 October 2026. Delivered title block “REPORT 119 / Inversion sequence asymptotics from two kernel roots / A279569 and class 1953A / 2 October 2026 / Self-contained proof and exact reproducibility companion”. Printed as delivered, with **[write]** notes. Delivered Section k is Section $k+9$ here, its statement $k.j$ is $(k+9).j$ and its equation (m) is $(m+72)$. The delivered “report118” is Part I; the delivered citation key `oeis` is `oeis1953`. Remark 10.2 and Subsection 19.1 are added in the write.

Reading conventions of Part II	Elsewhere in the report
$d_j = C_j/C$ relative, ℓ_j logarithmic, η_j inverse corrections	Part I: c_j, d_j, e_j ; Part III: d_j, ℓ_k, P_j
$e_j, c_j, d_j^{\text{orb}}$: orbit coefficients (87)	Part I: c_j relative corrections, e_j inverse coefficients
b_m : Puiseux coefficients; B_j Bernoulli polynomials; $B(L)$ in Section 17.2	Part I: b_n counts, $b(z, x)$ orbit forcing
Target T , $u(T)$, M_K , T_K ; $L = \log T$ in Section 17.2	Part I: Y , E_K , Y_K , $L = \log \mu$; Part III: M , $L = \log M$
α, β : the two fixed points of f_z ; $p = \frac{3}{2}$	Part I: $\beta = \frac{3}{2}$, α an exponent
$Q(z, X)$, $R(z, X)$ orbit limits; $M = 181/100$, $q = 7/25$, $P = 2^{230}$; $F(z, v, w)$	Part III: $R(z)$ ternary series, $P(z)$, $Q(z)$, $q = \frac{1}{2}$; $F(z)$ the generating function

Abstract of this Part (as delivered). For inversion sequences avoiding 110, 120 and 210, we prove

$$a_n \sim C \left(\frac{27}{4} \right)^n n^{-3/2}, \quad 0.01116841071267033797867 < C < 0.01116841071267033797869.$$

This establishes the leading law stated numerically by Britt and Beaton for class 1953A, counted by OEIS A279569. Two small roots of a cubic kernel generate contracting Möbius orbits. An exact cancellation gives jointly holomorphic orbit limits, and their even quotient identifies the combinatorial germ. A real determinant gap, Pringsheim’s theorem and removal of bounded boundary poles give a complex continuation domain. An outward rational certificate proves that the square-root amplitude does not vanish. We obtain a convergent local Puiseux expansion, coefficient asymptotics to every fixed algebraic order, certified first and second relative corrections, and Lambert- W_{-1} count-index inverses with shrinking two-ceiling brackets. The inverse remainder constants remain existential. The numerical amplitude was already recorded in the OEIS; no priority, nonalgebraicity, non- D -finiteness or external peer-review claim is made.

10 Model and main conclusions

10.1 The counted objects and source statement

An inversion sequence of length n is an integer vector $e = (e_1, \dots, e_n)$ with $0 \leq e_i < i$. Let a_n count those with no triple $i < j < k$ satisfying

$$e_i > e_k \quad \text{and} \quad e_j > e_k. \tag{73}$$

Equivalently, the reduced patterns 110, 120 and 210 are forbidden. Set $a_0 = 1$ and $A(z) = \sum_{n \geq 0} a_n z^n$. This is an ordinary generating function, with no factorial normalization or index shift. The sequence is OEIS A279569 [4]; its prefix is

$$1, 1, 2, 6, 22, 91, 409, 1953, 9763, 50583, 269697, 1472080.$$

Britt and Beaton [1, Section 3.7, equations (3.38)–(3.39)] give the exact succession rule and catalytic equation used below. Their label “1953A” uses the length-seven count. Section 4.2, equation (4.11), states the numerical prediction $a_n \sim C(27/4)^n n^{-3/2}$, with $C \approx 0.0111684$; the section explicitly treats its asymptotics as non-rigorous. We use version 3, dated 29 September 2026. The longer decimal amplitude in OEIS A279569 was entered by Václav Kotěšovec on 7 October 2021. The proof here establishes the law and certifies the amplitude; it does not claim discovery of those digits.

[write] 5 October 2026. Checked at placement against arXiv:2512.21943v3: Section 3.7 gives the rule and equations (3.38)–(3.39), and Section 4.2, equation (4.11), the law with $C \approx 0.0111684$, in the section whose asymptotics are declared non-rigorous. The OEIS value read at the write, 0.0111684107126703379786799829348... (Kotěšovec, 7 October 2021), agrees digit for digit with (77) and (115) as far as it is displayed. The certified intervals add certainty, not digits.

10.2 The theorem

Write $\rho = 4/27$ and $\lambda = \log(27/4)$.

Theorem 10.1. *The generating function A has radius of convergence ρ , and ρ is its unique singularity on the circle of convergence. For some $\eta > 0$ and $0 < \phi < \pi/2$, it extends holomorphically to the Δ -domain*

$$\{z : |z| < \rho + \eta, z \neq \rho, |\arg(z - \rho)| > \phi\}. \quad (74)$$

Near ρ it has a convergent expansion

$$A(z) = \sum_{m \geq 0} b_m t^m, \quad t = \sqrt{1 - z/\rho}, \quad b_1 < 0, \quad (75)$$

with the branch $t > 0$ for real $z < \rho$. For every fixed integer $J \geq 0$,

$$a_n = \left(\frac{27}{4}\right)^n n^{-3/2} \left(C_0 + \frac{C_1}{n} + \cdots + \frac{C_J}{n^J} + O(n^{-J-1})\right), \quad C_0 = C > 0. \quad (76)$$

The coefficients b_m, C_j are specified by convergent recurrences and explicit coefficient-transfer formulas below. In particular,

$$0.01116841071267033797867 < C < 0.01116841071267033797869. \quad (77)$$

For $N(T) = \min\{n \geq 0 : a_n \geq T\}$, Theorem 17.1 gives Lambert- W_{-1} expansions with an error retained inside two ceilings.

The constant in the remainder of (76) may depend on J . Convergence of the local series (75) does not assert convergence of the inverse-power expansion as $J \rightarrow \infty$. The inverse error constants and their starting thresholds are not numerically certified here.

10.3 How the continuation argument differs

This is a self-contained companion to report118, which treated classes 214 and 1509 using iterated kernels. The present representation needs both small cubic roots and a quotient of two orbit limits. In the class-1509 argument, a uniform denominator majorant rules out zeros on a complex domain. Here we prove a determinant gap only on the positive real segment. Normal convergence first gives meromorphic continuation around the rest of the critical circle; positivity then bounds the counting function up to that circle, forcing every possible boundary pole to be removable. This distinction matters: a real-axis nonvanishing proof alone would not justify singularity analysis.

[write] **5 October 2026.** “report118” is Part I; its classes 214 and 1509 are treated in Sections 1–7, and the “class-1509 argument” with its uniform denominator majorant is (49). Part III uses a third device for class 830, a zero-free orbit product on a larger disc (Proposition 24.1).

Remark 10.2 ([write], 5 October 2026). Britt and Beaton’s Table 1 [1] records that class 1953A is Wilf-equivalent to class 1953B, ($\neq, \geq, >$), that is the avoidance of 100, 120 and 210, so that both are counted by A279569; their Section 3.7 attributes the Wilf-equivalence to Martinez and Savage [8]. Hence every statement of this Part about the numbers a_n (Theorem 10.1 with its certified constants, Theorem 17.1, (128) and $a_{n+1} > a_n$ for $n \geq 1$) holds verbatim for the number of inversion sequences of length n having no $i < j < k$ with $e_i \neq e_j$, $e_j \geq e_k$ and $e_i > e_k$.

Proof. Wilf-equivalence means that the two classes have the same number of elements of every length, and the listed statements concern only these numbers. The generating tree, catalytic variables and orbits of this Part describe class 1953A, not 1953B; only the counting statements transfer. The Wilf-equivalence is Theorem 60 of Martinez and Savage [8] (arXiv:1609.08106v2, Section 3.2.1). Classes 1953A and 1953B are the inversion sequences avoiding $\{110, 210, 120\}$ and $\{100, 210, 120\}$, and Martinez and Savage prove the theorem by showing that their bijection between the sequences avoiding $\{110, 210\}$ and those avoiding $\{100, 210\}$ (their Section 2.27) preserves avoidance of 120, saying that the argument for its inverse is similar. As a check, a brute-force enumeration made at the write from the two definitions gives the same counts 1, 1, 2, 6, 22, 91, 409, 1953, 9763, 50583, 269697 for $n = 0, \dots, 10$, the displayed terms of A279569; and an independent dynamic program for class 1953B over the state (maximum, second distinct value, threshold), made at the independent check, agrees with one for class 1953A and with the A279569 b-file for every $n \leq 80$. (Dated note, 5 October 2026: the first sentence first read “The Wilf-equivalence is used as Britt and Beaton cite it (Martinez and Savage’s paper was not checked here)”; the paper was read at the independent check recorded at the end of Section 30.1.) $\square$

11 The exact succession rule and scalar equation

Every state at depth n of the generating tree contributes one object, with multiplicity. Start with the label $(p, s) = (0, 0)$. Its children are

$$\begin{aligned} (p, s) &\mapsto (p+1, s-i), & 0 \leq i \leq s, \\ &\mapsto (p+1-\ell, k), & 1 \leq \ell \leq p, \quad 0 \leq k < \ell. \end{aligned} \tag{78}$$

The labels record the lengths of the initial and possible second run of zeros in the source’s left-growing construction. Let $b_{n,p,s}$ be the number of labels, counting multiplicity, with $b_{0,p,s} = \mathbf{1}_{(p,s)=(0,0)}$. Rule

(78) determines $b_{n+1,p,s}$ by summing contributions of every child. Thus

$$B(z, x, y) = \sum_{n,p,s \geq 0} b_{n,p,s} z^n x^p y^s, \quad H(z, x) = B(z, x, 1), \quad A(z) = H(z, 1).$$

In particular each coefficient of z in H is a polynomial in x .

Summing the two finite child sums in (78) gives

$$\begin{aligned} B(z, x, y) &= 1 - \frac{zxy}{1-y} B(z, x, y) \\ &\quad - \frac{xz(x^2 + y - 2xy)}{(1-x)(x-y)(1-y)} H(z, x) \\ &\quad + \frac{zxy}{(1-y)(x-y)} H(z, y) + \frac{zx}{(1-x)(1-y)} A(z). \end{aligned} \tag{79}$$

The displayed rational denominators encode polynomial finite sums; identities at their exceptional values are interpreted after clearing denominators.

Define

$$\begin{aligned} f_z(x) &= \frac{1}{1-zx}, & K(z, x) &= (x-1)^2 - zx^3, \\ c(z, x) &= \frac{K(z, x)}{1-x}, & d(z, x) &= x - f_z(x), \\ e(z, x) &= 1 + \frac{zx^2}{1-x}. \end{aligned} \tag{80}$$

Multiplying (79) by $(1-y)(1-x)(x-y)$ makes its left kernel $(1-y+zx y)(1-x)(x-y)$. Set $y = f_z(x)$, cancel the kernel and simplify the remaining identity. The result is the scalar equation

$$H(z, f_z(x)) = c(z, x)H(z, x) + d(z, x) + e(z, x)A(z), \tag{81}$$

with the useful exact relation

$$c(z, x) = 1 - xe(z, x). \tag{82}$$

No analytic assumption on A is needed for this formal calculation.

12 Two roots and the formal counting germ

12.1 The two small kernel branches

Put $u = \sqrt{z}$. Let $X(u)$ be the unique formal solution of

$$X = 1 + uX^{3/2}, \quad X(0) = 1, \tag{83}$$

where the fractional power is expanded at 1. The implicit function theorem also makes this an analytic germ. Lagrange inversion gives

$$[u^n](X(u) - 1) = \frac{1}{n} \binom{3n/2}{n-1} > 0 \quad (n \geq 1). \tag{84}$$

The two roots of $K(z, x) = 0$ with finite limit at zero are $X_+(z) = X(u)$ and $X_-(z) = X(-u)$, beginning $1 + u + O(u^2)$ and $1 - u + O(u^2)$. At either root X ,

$$Y = f_z(X) = \frac{X^2}{2X-1}, \quad e(z, X) = \frac{1}{X}, \quad H(z, Y) = X - Y + \frac{A(z)}{X}. \tag{85}$$

These statements follow by replacing z with $(X-1)^2/X^3$ in (80) and using $c(z, X) = 0$.

12.2 The orbit and its canceled coefficients

For independent z, X define

$$x_{-1} = X, \quad x_0 = f_z(X), \quad x_{j+1} = f_z(x_j) \quad (j \geq 0). \quad (86)$$

The identity $x_j - 1 = zx_{j-1}x_j$ yields the cancellation

$$e_j := e(z, x_j) = 1 - \frac{x_j}{x_{j-1}}, \quad c_j = 1 - x_j e_j, \quad d_j^{\text{orb}} = x_j - x_{j+1}. \quad (87)$$

We use the right-hand side to define e_j at $z = 0$ as well. It removes the apparent division by $1 - x_j$. The superscript on d_j^{orb} distinguishes this orbit increment from the relative asymptotic corrections introduced later.

Initialize and iterate

$$\begin{aligned} Q_0 &= X - x_0, & R_0 &= 1/X, \\ Q_{j+1} &= c_j Q_j + d_j^{\text{orb}}, & R_{j+1} &= c_j R_j + e_j. \end{aligned} \quad (88)$$

Let $Q(z, X), R(z, X)$ be their limits. Section 13 proves normal convergence on an explicit product domain. The recurrence is also a definition by convergent product-sum expressions, but requires no c_j to be nonzero.

The attracting fixed point is

$$\alpha(z) = \frac{2}{1 + \sqrt{1 - 4z}}, \quad f_z(\alpha) = \alpha, \quad f'_z(\alpha) = \alpha - 1. \quad (89)$$

For the substitutions $z = u^2$, $X = X(\pm u)$, the exact difference formula

$$f_z(x) - f_z(y) = \frac{z(x - y)}{(1 - zx)(1 - zy)}$$

shows, coefficientwise and uniformly on a sufficiently small u -disk,

$$x_j - \alpha = O(u^{2j+3}), \quad e_j, \quad c_j - 1 = O(u^{2j+1}), \quad d_j^{\text{orb}} = O(u^{2j+3}). \quad (90)$$

Indeed $x_0 - \alpha = O(u^3)$ and each application of the difference formula gains a factor $O(u^2)$. These estimates make (88) coefficientwise convergent in $\mathbb{C}[[u]]$.

Proposition 12.1. *The formal counting series has the exact representation*

$$A(z) = \frac{Q(z, X_-(z)) - Q(z, X_+(z))}{R(z, X_+(z)) - R(z, X_-(z))}. \quad (91)$$

This quotient defines an analytic germ in z at zero, with value 1.

Proof. Start with (85) and iterate (81). Then $H(z, x_j) = Q_j + R_j A(z)$. Since each z -coefficient of H is polynomial in its second argument, the coefficientwise limit in (90) gives

$$H(z, \alpha) = Q(z, X_+) + R(z, X_+)A = Q(z, X_-) + R(z, X_-)A.$$

Solving these two equations gives (91). This argument uses the original formal counting series; it does not assume it already has a positive radius.

The joint holomorphy lemma below applies near $(z, X) = (0, 1)$. From (88) and (90),

$$Q(u^2, X(u)) = u + O(u^2), \quad R(u^2, X(u)) = 1 - u + O(u^2).$$

The numerator and denominator of (91) are both odd analytic functions of u , each with leading term $-2u$. Their quotient is even and holomorphic at $u = 0$, with value 1. An even holomorphic u -germ is holomorphic in $z = u^2$. It equals A coefficientwise by the formal identity, establishing positive radius of convergence. $\square$

13 An explicit joint holomorphy domain

Lemma 13.1. *The limits $Q(z, X), R(z, X)$ in (88) are jointly holomorphic on*

$$\mathcal{D} = \{(z, X) \in \mathbb{C}^2 : |z| < 3/20, \quad 0 < |X| < 31/10\}. \quad (92)$$

The recurrences converge normally on every compact subset of $\mathcal{D}$.

Proof. For $(z, X) \in \mathcal{D}$,

$$|x_0| < \frac{1}{1 - (3/20)(31/10)} = \frac{200}{107} < \frac{19}{10}.$$

The closed disk $|x| \leq 19/10$ is invariant under f_z , since

$$|f_z(x)| < \frac{200}{143} < \frac{19}{10}, \quad |f'_z(x)| < \frac{6000}{20449} < \frac{3}{10}. \quad (93)$$

Every orbit point is nonzero. Moreover,

$$|1/x_0| < \frac{293}{200}, \quad |1/x_j| < \frac{257}{200} \quad (j \geq 1). \quad (94)$$

The first bound follows from $1/x_0 = 1 - zX$, and the second from $1/x_j = 1 - zx_{j-1}$. The common bound $293/200$ is therefore valid even when the preceding point in (87) is x_0 .

Contraction gives, for $j \geq 1$,

$$|x_j - x_{j-1}| \leq (3/10)^{j-1} |x_1 - x_0|.$$

Consequently (87) and (94) make $e_j, c_j - 1, d_j^{\text{orb}}$ geometrically small. The coefficient e_0 uses $1/X$, which is uniformly bounded on each compact subset of $\mathcal{D}$. All finite recurrences are holomorphic there, including at $z = 0$.

If $\sum_j |c_j - 1|$ is bounded, every finite product of $|c_j|$ is bounded by $\exp(\sum_j |c_j - 1|)$. Formula (88) and the summable forcing terms then bound Q_j, R_j . The differences

$$Q_{j+1} - Q_j = (c_j - 1)Q_j + d_j^{\text{orb}}, \quad R_{j+1} - R_j = (c_j - 1)R_j + e_j$$

are uniformly summable on compact subsets. Thus Q_j, R_j converge normally and their limits are jointly holomorphic. The orbit itself converges to the fixed point (89), determined by $\alpha(0) = 1$. $\square$

Lemma 13.2. *Let $r = \sqrt{\rho} = 2/(3\sqrt{3})$. The series $X(u)$ has radius r , is absolutely convergent for $|u| \leq r$, and satisfies*

$$|X(u)| \leq 3 \quad (|u| \leq r), \quad X(r) = 3, \quad X(-r) = 3/4. \quad (95)$$

The relevant kernel branches extend analytically across each point $|z| = \rho$ other than $z = \rho$. At ρ they are analytic in $t = \sqrt{1 - z/\rho}$; the lower branch is analytic in t^2 .

Proof. Stirling's formula in (84) gives radius r and convergence at r . The positive real solution of (83) reaches $X = 3$ at $u = r$. Substitution at $u = -r$ gives $X = 3/4$ on the branch from 1. Positivity of the coefficients then proves the disk bound. No finite kernel root is zero, since $K(z, 0) = 1$.

The simultaneous equations $K = K_x = 0$ have only the nonzero critical value $z = 4/27$, with $x = 3$; the cubic discriminant is $z(4 - 27z)$. The double root at ρ has an ordinary square-root parametrization, while the other root $3/4$ is simple. Thus the implicit function theorem gives the asserted local continuations. $\square$

The strict inequalities $\rho < 3/20$ and $3 < 31/10$ place both selected seeds within the joint domain (92), with room to continue outside the critical circle. Hence (91) is meromorphic in a neighborhood of every noncritical boundary point, and meromorphic in t near ρ . This is a genuinely complex continuation statement, not a consequence of real-axis convergence.

14 The real determinant and removal of boundary poles

14.1 A strict determinant gap

For $0 < z \leq \rho$, the two real seeds obey

$$1 < X_+(z) \leq 3, \quad 3/4 \leq X_-(z) < 1.$$

The upper seed lies above α . A crossing is impossible because $K(z, \alpha) = -z\alpha^2 < 0$, and the ordering holds near zero. The fixed-point quadratic and monotonicity of f_z then give

$$\alpha < f_z(X_+) < X_+, \quad 1 < f_z(X_-) < \alpha.$$

The upper orbit decreases to α ; the lower orbit increases to it.

Lemma 14.1. *For $0 < z \leq \rho$,*

$$R(z, X_-(z)) \geq \frac{1}{X_-(z)} > 1 > R(z, X_+(z)). \quad (96)$$

In particular the denominator in (91) never vanishes on this positive segment, including at ρ .

Proof. On the upper orbit, $e_j > 0$ by (87). Also $0 < c_j < 1$: the kernel is negative between 1 and X_+ , so $c_j = K(z, x_j)/(1 - x_j) > 0$, while (82) gives $c_j < 1$. Starting from $0 < R_0 = 1/X_+ < 1$, the recurrence preserves positivity and

$$R_{j+1} - 1 = c_j(R_j - 1) + (1 - x_j)e_j < 0.$$

More quantitatively, $1 - R_{j+1} \geq c_j(1 - R_j)$. Since $\sum_j |1 - c_j| < \infty$ and every $c_j > 0$, the infinite product $\prod_j c_j$ is positive. Thus $1 - R \geq (1 - R_0) \prod_j c_j > 0$.

On the lower orbit, $x_j > 1$, $e_j < 0$ and $c_j > 1$. Starting from $R_0 = 1/X_- > 1$, induction gives $R_j > 1$ and

$$R_{j+1} - R_j = e_j(1 - x_j R_j) > 0.$$

Its limit is at least $1/X_- > 1$, proving (96). $\square$

The strict lower-branch gap alone would suffice: finite upper bounds $R_j < 1$ give $R \leq 1$ in the limit. The product argument above also establishes strictness of the upper limit; finite strict inequalities alone would not do so.

14.2 The radius and the boundary

Proposition 14.2. *The radius of A is at least ρ and*

$$\sum_{n \geq 0} a_n \rho^n = A(\rho) < \infty. \quad (97)$$

Every apparent pole of (91) on $|z| = \rho$, except possibly at ρ , is removable. The counting function has a Δ -domain of the form (74) and is holomorphic in t near ρ .

Proof. Proposition 12.1 gives a positive convergence radius. If it were less than ρ , Pringsheim's theorem for nonnegative coefficients would make its positive real endpoint singular. Lemmas 13.1 and 14.1, and the regular positive kernel branches before ρ , continue (91) holomorphically through that endpoint. This contradiction proves that the radius is at least ρ .

At ρ the orbit limits are finite and their determinant is nonzero, so the quotient has a finite radial limit. Since $a_n \geq 0$, monotone convergence gives (97). The series is therefore uniformly absolutely convergent on the closed disk, and $|A(z)| \leq A(\rho)$ there.

Fix ζ on the critical circle with $\zeta \neq \rho$. Lemma 13.2 and joint holomorphy provide a meromorphic quotient on a full neighborhood of ζ . Its denominator is not identically zero: on the u -sheet it is a single analytic function with expansion $-2u + O(u^3)$, and the identity theorem applies along the continued branches. The quotient agrees with A on the interior overlap by continuation of the germ. Any nonremovable singularity of this meromorphic function at ζ would be a pole, forcing unbounded values along the inward radius. That contradicts (97). The singularity is therefore removable.

At ρ , Lemma 14.1 makes the determinant nonzero, so the quotient is holomorphic in t . Small neighborhoods of the other circle points have holomorphic continuations after removal. A finite cover of the circle away from a small neighborhood of ρ , combined with the local slit neighborhood at ρ , gives some η, ϕ in (74). Identity on overlapping interior sets makes the continuations consistent. $\square$

Bounded radial values do not generally remove arbitrary singularities. The proof first establishes meromorphic continuation, where the only possible nonremovable isolated singularities are poles. It makes no claim that the orbit determinant has no complex zeros everywhere in (92); it shows that any relevant boundary zeros cancel in the quotient.

15 The positive amplitude and its exact certificate

15.1 The critical derivative

Treat the two seeds as independent variables and define near $(\rho, 3, 3/4)$

$$F(z, v, w) = \frac{Q(z, w) - Q(z, v)}{R(z, v) - R(z, w)}. \quad (98)$$

It is holomorphic there by Lemmas 13.1 and 14.1. The critical root expansion is

$$z = \rho(1 - t^2), \quad X_+ = 3 - 2\sqrt{3}t + O(t^2), \quad X_- = 3/4 + O(t^2). \quad (99)$$

Since z and X_- are analytic in t^2 , the coefficient of t is

$$b_1 = -2\sqrt{3}F_v(\rho, 3, 3/4), \quad C = \frac{-b_1}{2\sqrt{\pi}} = \sqrt{\frac{3}{\pi}}F_v(\rho, 3, 3/4). \quad (100)$$

It remains essential to prove $F_v > 0$; the kernel discriminant alone does not exclude cancellation of this term.

15.2 Uniform complex tail bounds

The following deliberately generous constants make a finite rational calculation an infinite-orbit certificate. Fix $z = \rho$ and let the seed X range over either of the closed disks

$$|X - 3| \leq h \quad \text{or} \quad |X - 3/4| \leq h, \quad h = 1/1000.$$

Set $M = 181/100$, $q = 7/25$ and $P = 2^{230}$. The rational inequalities

$$\frac{1}{1 - \rho(3 + h)} < M, \quad \frac{1}{1 - \rho M} < M, \quad \frac{\rho}{(1 - \rho M)^2} < q \quad (101)$$

show that all orbit points satisfy $|x_j| < M$ and that f_ρ is q -contracting on their disk. Lower bounds follow from

$$\begin{aligned} \frac{1}{|1 - \rho a| + \rho h} &> \frac{3}{4}, \quad \frac{\rho(a - h)}{1 + \rho(a + h)} > \frac{1}{12} \quad (a = 3, 3/4), \\ \frac{1}{1 + \rho M} &> \frac{3}{4}, \quad \frac{\rho(3/4)}{1 + \rho M} > \frac{1}{12}. \end{aligned} \quad (102)$$

The first line bounds $|x_0|$ and $|x_0 - 1|$ on the seed disks; the second line propagates $|x_j| > 3/4$ and $|x_j - 1| > 1/12$ for later points.

For the two fixed points α, β of f_ρ ,

$$1 < \alpha < 5/4, \quad \beta = \frac{1 + \sqrt{1 - 4\rho}}{2\rho} < 28/5, \quad |x_j - \alpha| < 4q^j.$$

The last estimate follows from $|x_0 - \alpha| < M + 5/4 < 4$ and contraction. Using

$$e(\rho, x) = -\frac{\rho(x - \alpha)(x - \beta)}{x - 1}, \quad \rho 12(M + 28/5) < 14,$$

we obtain

$$|e_j| < 56q^j, \quad |c_j - 1| < 104q^j, \quad |d_j^{\text{orb}}| < 6q^j. \quad (103)$$

Here $M \cdot 14 < 26$ and $4(1 + q) < 6$ suffice for the last two bounds. Since

$$\frac{104}{1 - q} < 145, \quad \exp(145) < 3^{145} < 2^{230} = P,$$

every finite product of $|c_j|$ is less than P . The initial values obey $|Q_0| < 5$, $|R_0| < 2$, so

$$|Q_j| < P \left(5 + \frac{6}{1 - q} \right) < 14P, \quad |R_j| < P \left(2 + \frac{56}{1 - q} \right) < 80P. \quad (104)$$

Summing consecutive differences in (88) from $j = N$ onward now gives

$$|Q - Q_N| < E_N, \quad |R - R_N| < E_N, \quad E_N = 2^{244} q^N. \quad (105)$$

Indeed both $(104 \cdot 14P + 6)/(1 - q)$ and $(104 \cdot 80P + 56)/(1 - q)$ are below 2^{244} . The error functions are holomorphic on the seed disks. Cauchy's estimate at their centers therefore bounds their first seed derivatives by $E_N/h = 1000E_N$.

15.3 Finite outward arithmetic

Use $N = 256$ in (88), and differentiate those recurrences with respect to the upper seed v . This can be done by dual-number arithmetic: for a value-derivative pair (a, a') , multiplication is $(a, a')(b, b') = (ab, a'b + ab')$ and inversion is $(a, a')^{-1} = (1/a, -a'/a^2)$. The initial upper pair is $(3, 1)$; the lower seed is $(3/4, 0)$, so its derivative with respect to v is exactly zero.

Every value is enclosed in a rational interval with endpoints on the grid $10^{-100}\mathbb{Z}$. Additions, multiplications and reciprocals are rounded outward by integer floor and ceiling; square roots use integer square roots. No floating-point estimate is an input. Before forming the quotient (98) or differentiating it, widen each value interval by E_{256} and each upper-seed derivative interval by $1000E_{256}$. The lower-seed derivative needs no widening.

For π , use Machin's formula

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239). \quad (106)$$

The alternating sum of the first m terms $\sum_{j=0}^{m-1} (-1)^j / ((2j+1)a^{2j+1})$ and that sum plus its next term enclose $\arctan(1/a)$. Taking $m = 100$ for $a = 5$ and $m = 30$ for $a = 239$ gives rational endpoints adequate for the claimed digits.

The included exact checker verifies all inequalities above and the finite recurrence, yielding in particular

$$0.01142893296759123231445 < F_v(\rho, 3, 3/4) < 0.01142893296759123231446. \quad (107)$$

Substitution in (100) proves (77). In particular $b_1 < 0$, so ρ is a genuine singularity. Combined with Proposition 14.2, this establishes the precise radius and unique dominant singularity. Agreement with previously posted decimals is a consistency check; the rational enclosure supplies the proof of noncancellation.

16 Every fixed algebraic correction

16.1 The convergent local series

The local kernel branches may be written explicitly as

$$\begin{aligned} s_+(t) &= \frac{1 + 2 \cos((\pi + 2 \arcsin t)/3)}{3}, & s_-(t) &= \frac{1 - 2 \cos(2 \arcsin(t)/3)}{3}, \\ X_+(t) &= \frac{1}{1 - s_+(t)}, & X_-(t) &= \frac{1}{1 - s_-(t)}. \end{aligned} \quad (108)$$

They satisfy $z = s^2(1 - s) = \rho(1 - t^2)$ and $K(z, X) = 0$; their values and first derivatives agree with (99). These are local analytic branch formulas. The orbit functions Q, R are still infinite normally convergent recurrences.

Define b_m by

$$F(\rho(1 - t^2), X_+(t), X_-(t)) = \sum_{m \geq 0} b_m t^m. \quad (109)$$

Joint holomorphy and the nonzero local determinant prove that this Taylor series converges for sufficiently small $|t|$. Arbitrarily many coefficients are obtainable by Taylor arithmetic in (88), with tails bounded using Cauchy's estimates on an appropriate fixed local domain.

16.2 Transfer coefficients and remainders

For $\gamma \notin \{0, 1, 2, \dots\}$, define $T_k(\gamma)$ by

$$\frac{\Gamma(n - \gamma)}{\Gamma(n + 1)} \sim n^{-\gamma-1} \sum_{k \geq 0} T_k(\gamma) n^{-k}. \quad (110)$$

They are obtained by exponentiating the formal series

$$\sum_{m \geq 1} \frac{(-1)^{m+1} [B_{m+1}(-\gamma) - B_{m+1}(1)]}{m(m+1)} n^{-m}, \quad (111)$$

where $B_j(x)$ are Bernoulli polynomials. In particular, $T_0 = 1$ and $T_1(\gamma) = \gamma(\gamma + 1)/2$. The exact coefficient identity

$$[z^n](1 - z/\rho)^\gamma = \rho^{-n} \frac{\Gamma(n - \gamma)}{\Gamma(-\gamma)\Gamma(n + 1)}$$

then gives

$$C_k = \sum_{j=0}^k \frac{b_{2j+1} T_{k-j}(j + 1/2)}{\Gamma(-j - 1/2)}. \quad (112)$$

For any specified J , expand (109) through t^{2J+4} with remainder $O(t^{2J+5})$. The local even powers form a polynomial in $1 - z/\rho$ and contribute no large- n algebraic terms. Transfer in the proved Δ -domain [6, Chapter VI] applies to the odd powers and the remainder. The last retained odd power t^{2J+3} contributes $O(\rho^{-n} n^{-J-5/2})$ and may be absorbed into the error in (76); the transferred local remainder is of still smaller order. Equations (110)–(112) prove (76) at every fixed depth. This completes the coefficient part of Theorem 10.1.

Let $d_j = C_j/C$ denote the relative corrections. The first two are specified exactly by

$$d_1 = \frac{3}{8} - \frac{3}{2} \frac{b_3}{b_1}, \quad d_2 = \frac{25}{128} - \frac{45}{16} \frac{b_3}{b_1} + \frac{15}{4} \frac{b_5}{b_1}. \quad (113)$$

These formulas are rigorous independently of a decimal evaluation of b_3, b_5 . The following certificate also supplies their first two numerical enclosures.

16.3 A certificate for the first two corrections

The constants in (113) admit the outward enclosures

$$\begin{aligned} 26.94860677861165 &< d_1 < 26.94860677861166, \\ 53.39335656811586 &< d_2 < 53.39335656811587. \end{aligned} \quad (114)$$

The same computation sharpens the amplitude enclosure to

$$\begin{aligned} 0.01116841071267033797867998293487861864 &< C, \\ C &< 0.01116841071267033797867998293487861865. \end{aligned} \quad (115)$$

The proof requires uniform control with varying z , rather than merely differentiation at $z = \rho$. We give the full error construction.

Set $h = 10^{-5}$ and $z(t) = \rho(1 - t^2)$. On $|t| \leq h$, put $w = (2/3) \arcsin t$. The integral formula for $\arcsin$ gives $|\arcsin t| \leq h/\sqrt{1 - h^2} < 3h/2$, hence $|w| < h$. The elementary power-series bounds $\cosh h < 2$, $|\cos w - 1| < h^2$ and $|\sin w| < 2h$ yield, from (108),

$$\begin{aligned} |s_+ - 2/3| &< 2h, & |X_+ - 3| &< \frac{18h}{1 - 6h} < 20h, \\ |s_- + 1/3| &< 2h^2/3, & |X_- - 3/4| &< \frac{3h^2/8}{1 - h^2/2} < h^2. \end{aligned} \quad (116)$$

Both seeds stay in the radius-1/1000 disks. The exact root jets through degree five are

$$\begin{aligned} X_+(t) &= 3 - 2\sqrt{3}t + \frac{10}{3}t^2 - \frac{53\sqrt{3}}{27}t^3 + \frac{272}{81}t^4 - \frac{1901\sqrt{3}}{972}t^5 + O(t^6), \\ X_-(t) &= \frac{3}{4} + \frac{1}{12}t^2 + \frac{11}{324}t^4 + O(t^6). \end{aligned} \quad (117)$$

They follow by substituting in $K(\rho(1 - t^2), X) = 0$. For the double branch one compares through degree six, fixing the first derivative $-2\sqrt{3}$; the simple lower branch is even.

For the uniform tail use

$$R_* = 149/1000, \quad M = 181/100, \quad q = 7/25, \quad z_* = \rho(1 - h^2), \quad a \in \{3, 3/4\}.$$

On the product of $|z - \rho| \leq \rho h^2$ and either seed disk $|X - a| \leq 1/1000$, we have $|z| < R_*$ and $|z| \geq z_*$. The following rational inequalities provide the orbit and reciprocal bounds used in the calculation:

$$\begin{aligned} \frac{1}{1 - R_*(3001/1000)} &< M, & \frac{1}{1 - R_*M} &< M, & \frac{R_*}{(1 - R_*M)^2} &< q, \\ \frac{1}{|1 - \rho a| + \rho h^2 a + R_*/1000} &> \frac{3}{4}, & \frac{z_*(a - 1/1000)}{1 + R_*(a + 1/1000)} &> \frac{1}{12}, \\ \frac{1}{1 + R_*M} &> \frac{3}{4}, & \frac{z_*(3/4)}{1 + R_*M} &> \frac{1}{12}. \end{aligned} \quad (118)$$

Thus $|x_j| < M$, $|x_j| > 3/4$, $|x_j - 1| > 1/12$ and contraction has ratio below q throughout this domain. Positivity of the Catalan series implies $|\alpha(z)| \leq \alpha(R_*) < 5/4$, where $R_*(5/4)^2 < 1/4$ proves the last inequality. For the other fixed point, $\beta(z) = 1/z - \alpha(z)$ gives $|\beta(z)| \leq 1/z_* + 5/4 < 9$. Consequently

$$|x_j - \alpha(z)| < 4q^j, \quad |e_j| < 80q^j, \quad |c_j - 1| < 148q^j, \quad |d_j^{\text{orb}}| < 6q^j.$$

The factorization of e used earlier still holds for these complex z . The numerical constants follow from $R_*12(M + 9) < 20$ and $M \cdot 20 < 37$.

Now $148/(1 - q) < 206$ and $\exp(206) < 3^{206} < 2^{327}$. Put $P_* = 2^{327}$. The same product-sum argument gives $|Q_j| < 14P_*$, $|R_j| < 114P_*$ and the uniform tail

$$|Q - Q_N|, |R - R_N| < \mathcal{E}_N, \quad \mathcal{E}_N = 2^{342}q^N. \quad (119)$$

For example, the bounding constants are justified by

$$5 + \frac{6}{1 - q} < 14, \quad 2 + \frac{80}{1 - q} < 114, \quad \frac{148 \cdot 14P_* + 6}{1 - q} < 2^{342}, \quad \frac{148 \cdot 114P_* + 80}{1 - q} < 2^{342}.$$

Use $N = 400$ from this point onward.

It remains to control division by the finite determinant throughout the t -disk. Write

$$D_N = R_N(z, X_+) - R_N(z, X_-), \quad U_N = Q_N(z, X_-) - Q_N(z, X_+), \quad F_N = U_N/D_N.$$

Represent a complex interval as a rectangle with real and imaginary rational interval components. Enclose z by the rectangle centered at ρ with component radii ρh^2 , X_+ by the rectangle centered at 3 with component radii $20h$, and X_- by the rectangle centered at $3/4$ with component radii h^2 . These rectangles contain the actual values for every $|t| \leq h$ by (116). Propagating them through the finite canceled recurrence (87)–(88), with outward arithmetic, gives

$$-2 < \Re D_N < -1, \quad |\Re U_N| + |\Im U_N| < 2. \quad (120)$$

This finite rational check proves $|D_N| > 1$ and $|F_N| < 2$ uniformly on the actual t -disk. Although the enclosing rectangles are larger than the actual image, (119) is applied only to actual $z(t), X_{\pm}(t)$ in its stated product domain.

By (119), the limiting numerator and denominator differ from U_N, D_N by less than $2\mathcal{E}_N$ each. Since $2\mathcal{E}_N < 1/2$, the limiting denominator has modulus greater than $1/2$. Subtracting the two quotients therefore gives

$$|F - F_N| < 12\mathcal{E}_N < 16\mathcal{E}_N \quad (|t| \leq h). \quad (121)$$

Indeed the numerator error divided by the limiting denominator is below $4\mathcal{E}_N$, and the denominator-error term is below $8\mathcal{E}_N$. Cauchy's coefficient estimate now proves

$$|b_j - [t^j]F_N| \leq 16\mathcal{E}_N h^{-j} \quad (j = 1, 3, 5). \quad (122)$$

Finite Taylor arithmetic uses (117), an outward enclosure of $\sqrt{3}$, and the convolution recurrence for reciprocal jets: if $a(t)b(t) = 1$, then $b_0 = 1/a_0$ and $b_n = -a_0^{-1} \sum_{k=1}^n a_k b_{n-k}$. Widening the resulting coefficients by (122), and substituting into (100) and (113), proves (114) and (115). The companion stores tighter rational endpoints and checks every displayed bracket. This certificate establishes the specified constants, not numerical remainder constants or starting indices for (76) or (127).

17 Counting index inverses and integer thresholds

17.1 Monotonicity and the Lambert approximation

Appending the new maximum n to a length- n inversion sequence preserves (73): the new entry can only be the last member of a triple, and no preceding value exceeds it. This is an injection into the size- $n + 1$ class. For $n \geq 1$, the all-zero sequence is outside its image, so $a_{n+1} > a_n$. Thus the threshold $N(T)$ is well defined for all sufficiently large T .

Write $p = 3/2$, $\lambda = \log(27/4)$ and define

$$u(T) = -\frac{p}{\lambda} W_{-1} \left(-\frac{\lambda}{p} \left(\frac{C}{T} \right)^{1/p} \right). \quad (123)$$

Here W_{-1} is the real Lambert branch with values at most -1 [7]. For large T , u is the large positive solution of $C \exp(\lambda u) u^{-p} = T$, and $u \rightarrow \infty$.

Distinguish the relative corrections d_j from the logarithmic corrections ℓ_j defined formally by

$$\log \left(1 + \sum_{j \geq 1} d_j n^{-j} \right) = \sum_{j \geq 1} \ell_j n^{-j}. \quad (124)$$

In particular $\ell_1 = d_1$, $\ell_2 = d_2 - d_1^2/2$. Define η_j recursively by cancellation in

$$0 = \lambda(v - u) - p \log(v/u) + \sum_{j \geq 1} \ell_j v^{-j}, \quad v = u + \sum_{j \geq 1} \eta_j u^{-j}. \quad (125)$$

The coefficient of the new unknown η_j is always λ , so the recursion determines every fixed order uniquely. Its first terms are

$$\eta_1 = -\frac{\ell_1}{\lambda}, \quad \eta_2 = -\frac{p\ell_1}{\lambda^2} - \frac{\ell_2}{\lambda}. \quad (126)$$

Theorem 17.1. *For any fixed $K \geq 0$, set $v_K(T) = u(T) + \sum_{j=1}^K \eta_j u(T)^{-j}$. There exist constants $M_K > 0$ and T_K such that, for every real $T \geq T_K$,*

$$\left\lceil v_K(T) - M_K u(T)^{-K-1} \right\rceil \leq N(T) \leq \left\lceil v_K(T) + M_K u(T)^{-K-1} \right\rceil. \quad (127)$$

Equivalently, some $\varepsilon_K(T) = O(u(T)^{-K-1})$ satisfies $N(T) = \lceil v_K(T) + \varepsilon_K(T) \rceil$. The two integers in (127) are eventually equal or adjacent.

Proof. The coefficient expansion and $C > 0$ give, at integer n ,

$$\log a_n = \lambda n - p \log n + \log C + \sum_{j=1}^K \ell_j n^{-j} + O(n^{-K-1}).$$

Let $G_K(x) = \lambda x - p \log x + \log C + \sum_{j=1}^K \ell_j x^{-j}$ for real x large. Some constant D_K makes $G_K(n) - D_K n^{-K-1} \leq \log a_n \leq G_K(n) + D_K n^{-K-1}$ at all sufficiently large integers. Both smooth envelopes have derivative $\lambda - p/x + O(x^{-2}) > \lambda/2$ for large x . This differentiates explicit finite smooth expressions, not the unspecified remainder of an integer sequence.

Formal cancellation (125), used at a fixed finite order, gives $G_K(v_K) - \log T = O(u^{-K-1})$. The mean value theorem puts the inverse roots of both envelopes at $v_K + O(u^{-K-1})$. An integer strictly below the inverse root of the upper envelope has count below T ; an integer at or above the inverse root of the lower envelope has count at least T . Taking ceilings and enlarging M_K gives (127). Finite initial counts are below T once T is sufficiently large. The real bracket width tends to zero, so its two ceilings are eventually equal or adjacent. $\square$

The unknown constants M_K, T_K are existential. The theorem is not an effective certified algorithm for a prescribed finite target. Dropping the error inside the ceiling is invalid uniformly over all real T : a target arbitrarily close to an exact count can change the last integer.

[write] **5 October 2026.** This is Part I's Theorem 7.1 for A279569: Part I's

$$(\mu, C, c_j, d_j, e_j, Y, y_K, E_K, Y_K, L, \beta) \quad \text{are} \quad (27/4, C, d_j, \ell_j, \eta_j, T, v_K, M_K, T_K, \lambda, p)$$

here, (126) is (71) under this dictionary, and the proof is a second write-up of the same envelope argument. Like Part I's, it is an instance of the Lambert-core, staircase and forward-to-inverse results of the transseries volume [9]; see the note after Theorem 7.1.

17.2 An explicit logarithmic version

Put $L = \log T$ and $B(L) = p \log L - p \log \lambda - \log C$. Expanding the same inverse gives a real crossing approximation

$$s(T) = \frac{1}{\lambda} \left[L + B(L) + \frac{pB(L) - \lambda d_1}{L} + O\left(\frac{(\log L)^2}{L^2}\right) \right]. \quad (128)$$

At each further fixed order the correction is a polynomial in $\log L$ divided by a power of L , determined by substitution in

$$L = \lambda n - p \log n + \log C + \log(1 + d_1/n + d_2/n^2 + \cdots).$$

Equation (128) is a continuous approximation; the rounding qualification in Theorem 17.1 remains necessary.

[write] **5 October 2026.** Part III's (136) is the same expansion for A279558, stated independently: there $A = \log C + p \log \lambda$, so $B(L) = p \log L - A$, which Part III calls q_L , and $pB(L) - \lambda d_1 = p^2 \log L - pA - \lambda d_1$ is its $P_1(\log L)$ (192). The error term of (128), stated here after "Expanding the same inverse", is the residual estimate proved in Part III's Proposition 28.1: its proof places the inverse roots of both envelopes within $\varepsilon_1(L) = B_1(1 + \log L)^2/L^2$ of $\nu_1(L)$, and $\nu_1(L)$ is $s(T)$ without its error term. Its argument uses only the form (188) of the coefficient expansion, which Theorem 10.1 supplies for A279569.

18 Reproducibility and scope

The companion includes the standalone mathematical source, rendered PDF, portable exact verification code and fixtures, and deterministic build and archive scripts. The exact model checks distinguish the 26 displayed OEIS terms through $n = 25$ from the 61 coefficients $n = 0, \dots, 60$ independently generated by both the state recurrence and orbit-series reconstruction. Brute-force inversion-sequence counts agree through $n = 8$, and scalar-functional-equation checks run through degree 16 at two rational catalytic arguments. These supply finite consistency checks. They do not replace the proofs of normal convergence, continuation and coefficient transfer.

The amplitude certificate consists of the finite interval recurrence, the proved uniform tail (105), derivative widening by Cauchy's estimate, the Machin bounds (106) and exact square roots. The correction certificate additionally uses the varying- z bound (119), the complex determinant check (120), and Cauchy coefficient errors (122). All interval endpoints and displayed strict inequalities can be checked using rational arithmetic alone. The separate checker README gives its exact schema, executed checks, and mutation scope.

The top-level inventory is closed: changed, missing and unexpected files are rejected. Checksums detect changes; they are not authenticated signatures. Build reproducibility means byte identity in the tested toolchain across fresh directories, not identity under arbitrary TeX, font, Python or compression versions. No source-paper PDFs, private research notes or rendering intermediates are needed in the package.

The theorem gives all fixed algebraic orders and a convergent local Puiseux expansion. It does not establish a convergent large- n inverse-power series, a complete exponentially improved expansion, nonalgebraicity or non- D -finiteness. A local square-root singularity is compatible with either algebraic or nonalgebraic behavior. No external peer review or global literature priority is asserted.

[write] **5 October 2026.** In this repository the companion is shipped under the prefix `119-tworoot` in `code/` and `data/`, with `119-tworoot-checks-README.md`; its `integrity.py`, a byte copy of

Report 118's, is not shipped separately (`code/118-kernels-integrity.py` is the same file), and the PDF, `CHECKSUMS.sha256` and `checks/MANIFEST.json` are not shipped. “The separate checker README” is `119-tworoot-checks-README.md`. The checks are rerun from the archive; see the report's README.

19 Further research questions

1. **Effective coefficient and inverse errors.** Can the qualitative Δ -domain be replaced by an explicit domain and usable bounds on it, yielding certified M_K, T_K in (127)? This would turn the asymptotic inverse into a finite-target procedure requiring at most a small final exact search.
2. **Complex determinant zeros.** Where do the zeros of $R(z, X_+) - R(z, X_-)$ lie beyond the critical circle, and which cancel with its numerator? The present proof removes possible boundary poles but does not classify farther zeros.
3. **Higher coefficients and late-order behavior.** Can efficiently certified Taylor jets expose the sign pattern and large-order growth of C_j ? Fixed-order existence does not determine convergence, optimal truncation or exponentially small corrections.
4. **Global algebraic and differential status.** Can the orbit representation prove or disprove algebraicity or D -finiteness? This requires information beyond the single dominant square-root expansion established here.
5. **Other two-root catalytic models.** Which other avoidance classes permit a real determinant gap and nonnegative boundary convergence in place of a global denominator majorant? Each model still needs proofs of formal-germ identification, complex continuation and noncancellation.

19.1 Further questions and research

[write] (5 October 2026.) Under Vladimir's standing rule of 4 October 2026, this subsection collects every claim of Part II that is stated without proof or left open, with its source, the sketch offered and what is missing. Nothing in this Part was found to be wrong, and no claim was refuted.

1. *Effective coefficient and inverse errors* (question 1; the paragraphs after Theorems 10.1 and 17.1). Sketch: an explicit Δ -domain with usable bounds. Missing: those bounds, explicit transfer constants and certified M_K, T_K ; as Part I, item 1 of Section 9.1.
2. *Complex determinant zeros* (question 2; the paragraph after Proposition 14.2). The proof claims no zero-free region for $R(z, X_+) - R(z, X_-)$ off the positive segment; it shows only that boundary zeros cancel. Missing: the location of the zeros beyond the critical circle and their cancellation with the numerator.
3. *Higher coefficients and late-order behaviour* (question 3). The C_j are specified by (112) for every j , but only d_1, d_2 are certified (114). Open: their sign pattern and growth, convergence or divergence of the $1/n$ series, optimal truncation and exponentially small corrections.
4. *Algebraic and differential status* (question 4). Source: Britt and Beaton write of class 1953A that they “do not believe it to be algebraic” (their Section 3.7). Missing: information beyond the single dominant square-root expansion.

5. *Other two-root catalytic models* (question 5). Each needs its own formal-germ identification, complex continuation and noncancellation proof.
6. *Class 1953B*. Remark 10.2 transfers the counting statements through a Wilf-equivalence proved by Martinez and Savage (their Theorem 60, by an explicit bijection that preserves avoidance of 120) and checked here for $n \leq 80$. A direct treatment of 1953B, or a bijection carrying the tree of this Part to it, is not given: Martinez and Savage’s bijection acts on sequences and is not shown to carry this Part’s generating tree. (Dated note, 5 October 2026: the first sentence first read “... through a Wilf-equivalence cited from Martinez and Savage and not re-checked here beyond $n \leq 10$ ”.)
7. *Earlier rigorous proofs*. No priority is claimed; the wider literature was not searched (as Part I, item 6).

Part III

A279558 asymptotics and inversion

Bundle Report 122, archive `A279558_Asymptotic_Expansion_and_Inverse_Thresholds_Source.zip`, delivered as `report122.tex` (17 pages), dated 2 October 2026. Delivered title block “REPORT 122 / A279558 asymptotics and inversion / A ternary root and two contracting orbits / 2 October 2026 / Standalone proof and exact reproducibility companion”. Printed as delivered, with `[write]` notes. Delivered Section k is Section $k + 19$ here, its statement $k.j$ is $(k + 19).j$ and its equation (m) is $(m + 128)$. The delivered citation key `oeis` is `oeis830`. Remark 28.2 and Subsection 30.1 are added in the write.

Reading conventions of Part III	Elsewhere in the report
$F(z)$: the generating function; $A = \log C + p \log \lambda$, a constant	Parts I, II: $A(z)$ the generating function
$(a, b)_S$, $(a, b)_T$: tree labels and types; S , T their series	a_n counts (all Parts); Part II: T a target
d_j relative, ℓ_k logarithmic corrections; $P_j(\log L)$ inverse polynomials; $c(z, x)$, $d(z, x)$ functions (147)	Part I: c_j , d_j (logarithmic), e_j ; Part II: ℓ_j , η_j
Target M , $L = \log M$; constants M_* , B_* , Z , K_* , L_* , K_0 , E_N	Part I: Y , $L = \log \mu$; Part II: T , $M = 181/100$
$R(z)$: the ternary series; $P(z)$, $Q(z)$, $V(z, R)$ orbit limits; $G(z, R)$ their quotient	Part II: $Q(z, X)$, $R(z, X)$ orbit limits; Part I: $P(z, x)$ a kernel polynomial
b_j : Puiseux coefficients; g_k : gamma-ratio polynomials	Part I: h_j , g_j ; Part II: b_m , T_k

Abstract of this Part (as delivered). Let a_n count inversion sequences of length n avoiding the patterns 010, 120, and 210. We prove

$$a_n \sim C \left(\frac{27}{4} \right)^n n^{-3/2}, \quad C = 0.0001809637754651179072704308064957871691430 \dots > 0.$$

The proof transforms the published generating tree into an affine functional equation. An algebraic anchor from $R = 1 + zR^3$ and two contracting rational orbits give an exact analytic representation. A zero-free denominator on $|z| \leq 149/1000$ excludes competing poles, and a certified nonzero square-root coefficient establishes the unique dominant singularity $4/27$. A convergent local Puiseux series yields coefficient asymptotics to every fixed order, including certified first relative corrections $137.22155058 \dots$ and $4914.56133466 \dots$. We also prove all-order logarithmic inversion and retain the necessary integer-rounding uncertainty at threshold crossings. Exact directed interval calculations have explicit infinite-tail error bounds. The leading numerical amplitude was previously credited to Václav Kotěšovec; the result here supplies a proof and a controlled all-fixed-order extension, rather than a claim of new numerical digits.

20 Definitions and principal results

An inversion sequence of length n is an integer sequence $e = (e_1, \dots, e_n)$ satisfying $0 \leq e_i < i$. A pattern occurs when a subsequence has the indicated order and equality relations. Avoidance of 010, 120, and 210 is equivalent to the absence of indices $i < j < k$ for which

$$e_i \neq e_j, \quad e_j > e_k, \quad e_i \geq e_k. \quad (129)$$

Indeed, if $e_i = e_k$ the pattern is 010; otherwise the two possible orders of the distinct first two entries give 120 or 210. Let a_n count these sequences, include the empty sequence by $a_0 = 1$, and put

$$F(z) = \sum_{n \geq 0} a_n z^n, \quad \rho = \frac{4}{27}, \quad \lambda = \log \frac{27}{4}, \quad p = \frac{3}{2}. \quad (130)$$

This is OEIS A279558 and Class 830 in Britt–Beaton [1, 5]. Its first terms are

$$1, 1, 2, 5, 15, 52, 200, 830, 3654, 16869, 80963, 401300, \dots$$

Theorem 20.1. *The generating function F has radius of convergence ρ and a unique singularity on $|z| = \rho$, namely $z = \rho$. It has analytic continuation to a Δ -domain there and a convergent local expansion*

$$F(z) = \sum_{j \geq 0} b_j \left(1 - \frac{z}{\rho}\right)^{j/2}, \quad b_1 < 0. \quad (131)$$

For every fixed integer $J \geq 0$,

$$a_n = \rho^{-n} n^{-3/2} \left(\sum_{j=0}^J \frac{C_j}{n^j} + O_J(n^{-J-1}) \right), \quad C_0 = C = -\frac{b_1}{2\sqrt{\pi}} > 0. \quad (132)$$

All coefficients are computable from a normally convergent orbit representation. In particular, with $d_j = C_j/C$,

$$\begin{aligned} C &> 0.0001809637754651179072704308064957871691430, \\ C &< 0.0001809637754651179072704308064957871691431. \end{aligned} \quad (133)$$

$$137.22155058455044742556 < d_1 < 137.22155058455044742557, \quad (134)$$

$$4914.56133466923234229799 < d_2 < 4914.56133466923234229800. \quad (135)$$

Here a Δ -domain can be taken to have the form

$$\{z : |z| < \rho + \eta, \ z \neq \rho, \ |\arg(z - \rho)| > \theta\}, \quad \eta > 0, \quad 0 < \theta < \pi/2.$$

Only the existence of suitable η, θ is needed. We do not claim an explicit numerical threshold n_0 for the asymptotic remainders. The series in (131) converges locally. The infinite expansion in powers of $1/n$ represented by (132) is an asymptotic expansion; its convergence is not asserted.

Theorem 20.2. *For $M > 0$ define the integer threshold $N(M) = \min\{n \geq 0 : a_n \geq M\}$. Put $L = \log M$ and $A = \log C + p \log \lambda$. The smooth asymptotic inverse begins*

$$\nu_1(L) = \frac{1}{\lambda} \left[L + p \log L - A + \frac{p^2 \log L - pA - \lambda d_1}{L} \right]. \quad (136)$$

There are a constant $B > 0$ and a sufficiently large L_0 such that, for $L \geq L_0$, with $\varepsilon_1(L) = B(\log L)^2/L^2$,

$$[\nu_1(L) - \varepsilon_1(L)] \leq N(e^L) \leq \lceil \nu_1(L) + \varepsilon_1(L) \rceil. \quad (137)$$

An analogous statement holds to every fixed order, with recursively computable polynomials in $\log L$ and an error tending to zero. If the two ceilings agree, their common value is the exact threshold. If they do not, rounding uncertainty remains.

[write] **5 October 2026.** Theorem 20.2 is the case $J = 1$ of Proposition 28.1, proved in Section 28. Indeed ν_1 of (136) is ν_J of (194) at $J = 1$, since $P_1(\log L) = pq_L - \lambda d_1 = p^2 \log L - pA - \lambda d_1$ by (192). Proposition 28.1 gives (195) with its error at $J = 1$, written here $\varepsilon_1^P(L) = B_1(1 + \log L)^2/L^2$ to distinguish it from the theorem's ε_1 , for $L \geq L_1$. For $L \geq e$ one has $1 + \log L \leq 2 \log L$, so $\varepsilon_1^P(L) \leq 4B_1(\log L)^2/L^2$, and enlarging ε only lowers the left ceiling and raises the right one. Hence (137) holds with $B = 4B_1$ and $L_0 = \max(L_1, e)$. This is the same theorem as Part I's Theorem 7.1 and Part II's Theorem 17.1, in the coordinate $L = \log M$; see the note after Proposition 28.1.

20.1 Published inputs and the scope of the result

Britt–Beaton [1, Section 2.4, equations (2.40)–(2.43)] give the exact generating tree and functional equations for this class. Their version 3, dated 29 September 2026, treats its asymptotics numerically in Section 4.2 and Appendix B.1. We rederive the needed tree and transformed equations below so the proof can be read independently.

The OEIS formula credits Václav Kotěšovec, 13 January 2026, with the numerical constant $c = 0.000320749940697549374489469232116484176 \dots$ in the normalization $c/\sqrt{\pi}$. Our constant is $C = c/\sqrt{\pi}$. The agreement of numerical digits is therefore a consistency check, not a new numerical discovery. The contribution established here is an exact analytic representation, a proof of the leading law, and a controlled all-fixed-order extension with inversion. No claim of universal literature priority is made. In particular, neither nonalgebraicity nor non-D-finiteness of F follows from this report.

[write] **5 October 2026.** Checked at placement against arXiv:2512.21943v3: Section 2.4 gives the class (“we do not believe this class to have an algebraic generating function”) with equations (2.40)–(2.43); Section 4.2, equation (4.9), states $C \approx 0.00018$, and Appendix B.1, equation (B.5), $C = 0.0001809637754651179$, both numerical. At the write the OEIS constant $c = 0.000320749940697549374489469232116484176 \dots$ (Kotěšovec, 13 January 2026) was compared with (133): c agrees digit for digit, as far as it is displayed, with $C\sqrt{\pi}$ for every C in that interval. The re-derived tree of Section 21 agrees with brute force from (129) for $n \leq 9$ at placement and $n \leq 10$ at the write, and the shipped table `data/122-anchor-a279558_n0_150.txt` with an independent dynamic program through $n = 150$ (placement).

21 The generating tree and the all zero correction

For a nonempty sequence let h be its maximum and k its greatest entry strictly below h , taking $h = k = 0$ for an all-zero sequence. Every sequence in the class ends on its maximum or its premaximum. To see this, if its final entry v were below two earlier distinct values h, k , those two values, in either order, followed by v would form 120 or 210.

Use type S when the last entry is h , and type T when the last entry is $k < h$. There is no T type among all-zero sequences. Set

$$a = n - h, \quad b = h - k. \tag{138}$$

The empty root is $(0, 0)_S$. From either type, appending the current maximum gives $(a + 1, b)_S$; appending a new maximum $h + d$, for $1 \leq d \leq n - h = a$, gives $(a + 1 - d, d)_S$.

The possible smaller appended values depend on the type. From S , a new value $v < h$ must satisfy $k < v < h$. A value below k would produce a forbidden triple using k and h ; a value equal to k would produce 010 using an earlier k and the final h . Conversely, when $k < v < h$, every earlier value below h is strictly below v , so a triple ending at v cannot satisfy (129). The new premaximum

is v . From T , the value k itself may also be appended: repeating the final value cannot create a new forbidden triple that was absent when this same final value was first present. Values $k < v < h$ are again valid, and values below k are not. Thus the complete rule is

$$(a, b)_{S \text{ or } T} \longrightarrow (a + 1, b)_S, \quad (139)$$

$$(a, b)_{S \text{ or } T} \longrightarrow (a + 1 - d, d)_S, \quad 1 \leq d \leq a, \quad (140)$$

$$(a, b)_S \longrightarrow (a + 1, j)_T, \quad 1 \leq j \leq b - 1, \quad (141)$$

$$(a, b)_T \longrightarrow (a + 1, j)_T, \quad 1 \leq j \leq b. \quad (142)$$

An empty index range means no children. In particular, (141) produces none when $b = 0$.

Let $S(z, x, y)$ and $T(z, x, y)$ mark size by z and labels a, b by x, y , respectively. Write

$$B(z, x, y) = S(z, x, y) + T(z, x, y), \quad H(z, x) = B(z, x, 1), \quad F(z) = H(z, 1).$$

When z is fixed in notation, $H(x)$ means $H(z, x)$, and similarly for B, S, T . Coefficients of each power of z are polynomials in x, y , so the divided differences below are formal polynomial identities.

Proposition 21.1. *The generating tree gives*

$$S = 1 + zx B + \frac{zxy}{x - y}(H(x) - H(y)), \quad (143)$$

$$(1 - zx)T = \frac{zx}{1 - y}(yH(x) - B) + \frac{zx}{1 - zx}. \quad (144)$$

Proof. For new maxima, the finite sum for a parent with first label a is

$$\sum_{d=1}^a x^{a+1-d} y^d = \frac{xy}{x - y}(x^a - y^a),$$

which gives (143). For T children, the two sums are

$$\sum_{j=1}^{b-1} y^j = \frac{y - y^b}{1 - y} \quad (b \geq 1), \quad \sum_{j=1}^b y^j = \frac{y(1 - y^b)}{1 - y}.$$

The first rational expression would equal -1 at $b = 0$, whereas the correct empty sum is zero. Exactly the all-zero sequences have type S with $b = 0$, and their generating function is

$$S(z, x, 0) = \sum_{n \geq 0} (zx)^n = \frac{1}{1 - zx}.$$

Consequently

$$T = \frac{zx}{1 - y}(yH(x) - S - yT) + \frac{zx}{1 - zx}.$$

Using $S + yT = B - (1 - y)T$ proves (144). □

The final term in (144) is essential. Omitting it changes both the exact enumeration and the later scalar equation.

22 A scalar equation and an algebraic anchor

Eliminate $T = B - S$ from (143)–(144). After multiplication by $1 - y$, the coefficient of B is

$$\mathcal{K}(z, x, y) = (1 - zx)^2(1 - y) + zx.$$

Its formal root in y is the rational map

$$\phi_z(x) = 1 + \frac{zx}{(1 - zx)^2}. \quad (145)$$

Substitution of $y = \phi_z(x)$ is valid coefficientwise. The remaining terms simplify to the one-variable affine equation

$$H(z, \phi_z(x)) = c(z, x)H(z, x) + d(z, x), \quad (146)$$

where

$$d(z, x) = \frac{x - \phi_z(x)}{zx}, \quad c(z, x) = 1 - (1 - zx)d(z, x). \quad (147)$$

For a general indeterminate x , equation (146) is read after clearing denominators. The apparent negative powers of z cancel in that identity. Along the orbits used below they are removable singularities; this will be proved explicitly.

Let $R(z)$ be the unique formal series satisfying

$$R = 1 + zR^3, \quad R(0) = 1. \quad (148)$$

Writing $R = 1 + W$ and applying Lagrange inversion gives

$$[z^n]R(z) = \frac{1}{2n+1} \binom{3n}{n} \quad (n \geq 0). \quad (149)$$

Put $X = R^2$. Since $zR^2 = (R - 1)/R$, direct algebra gives

$$\phi_z(X) = R^2 - R + 1 =: Y, \quad d(z, X) = R, \quad c(z, X) = 0. \quad (150)$$

For instance, $X - Y = R - 1$ and $zX = (R - 1)/R$ yield $d = R$; then $1 - zX = 1/R$ yields $c = 0$. Equation (146) therefore has the exact anchor

$$H(z, Y) = R. \quad (151)$$

22.1 The two orbit construction

For analytic purposes, regard z and R as independent variables. The upper orbit carries the anchor value:

$$\begin{aligned} y_0 &= \phi_z(R^2), & v_0 &= R, \\ y_{j+1} &= \phi_z(y_j), & v_{j+1} &= c(z, y_j)v_j + d(z, y_j). \end{aligned} \quad (152)$$

The lower orbit starts at the desired specialization $x = 1$:

$$\begin{aligned} x_0 &= 1, & p_0 &= 1, & q_0 &= 0, \\ x_{j+1} &= \phi_z(x_j), & p_{j+1} &= c(z, x_j)p_j, & q_{j+1} &= c(z, x_j)q_j + d(z, x_j). \end{aligned} \quad (153)$$

Denote the limits, when established, by

$$V(z, R) = \lim v_j, \quad P(z) = \lim p_j, \quad Q(z) = \lim q_j.$$

Only V depends on R . The solution will be

$$G(z, R) = \frac{V(z, R) - Q(z)}{P(z)}, \quad F(z) = G(z, R(z)). \quad (154)$$

Lemma 22.1. *Equation (154) holds as an identity of formal power series.*

Proof. The exact difference identity is

$$\phi_z(x) - \phi_z(y) = \frac{z(x-y)(1-z^2xy)}{(1-zx)^2(1-zy)^2}. \quad (155)$$

For $x(0) = y(0) = 1$, this increases the z -adic order of their difference by at least one. Each orbit therefore converges coefficientwise to the same formal fixed point $\alpha = 1 + O(z)$. On setting $R = R(z)$, equations (151), (146) give

$$v_j = H(z, y_j), \quad p_j F(z) + q_j = H(z, x_j).$$

Since every coefficient of $H(z, x)$ is a polynomial in x , passage to the common formal limit is legitimate. The lower first step has $d(z, 1) = -(1-z)^{-2}$ and $c(z, 1) = (2-z)/(1-z)$. All subsequent c factors have constant term 1 and their d terms have constant term zero. Thus $P(0) = 2$, $Q(0) = -1$. At $R = 1, z = 0$, the upper limit is 1, so $G(0, 1) = 1$. In particular P is a formal unit and (154) follows. $\square$

The finite quotient $(v_m - q_m)/p_m$ is generally not equal to F . As a formal approximation on the algebraic branch, however,

$$\frac{v_m - q_m}{p_m} - F = O(z^{m+2}). \quad (156)$$

Indeed $y_0 - x_0 = O(z)$ and (155) yields $y_m - x_m = O(z^{m+1})$, whereas $H(z, x) = 1 + O(z)$ has $\partial_x H = O(z)$. Subtract the two displayed identities in the proof and divide by the unit p_m .

23 Normal convergence and joint holomorphy

Proposition 23.1. *The functions $V(z, R)$, $P(z)$ and $Q(z)$ are holomorphic on*

$$|z| < \frac{3}{20}, \quad |R| < \frac{151}{100}. \quad (157)$$

The finite orbit expressions converge normally there.

Proof. For the upper seed, $|R^2| < 23/10$, and hence

$$|\phi_z(R^2) - 1| < \frac{69/200}{(131/200)^2} < \frac{81}{100}.$$

Both initial orbit points lie in $|x| \leq 181/100$. On this disk,

$$|\partial_x \phi_z(x)| \leq \frac{(3/20)(1 + (3/20)(181/100))}{(1 - (3/20)(181/100))^3} < \frac{1}{2}, \quad (158)$$

$$|\phi_z(x) - 1| < \frac{52}{100}. \quad (159)$$

The disk is invariant and convex, so the derivative bound gives a uniform contraction. It yields a common holomorphic fixed point $\alpha(z)$ and geometrically decaying consecutive differences. All orbit arguments are nonzero: the upper initial point has modulus greater than $19/100$, and all its subsequent points have modulus greater than $48/100$. The lower initial point is 1 and its subsequent points satisfy the latter bound as well. Also the factors $1 - zx$ stay away from zero.

It remains to remove the apparent $1/z$ in (147). The lower initial step has the explicit regular expression already given. For an upper initial point $u = \phi_z(S)$, one has the regular identity

$$d(z, u) = \frac{S/(1 - zS)^2 - u/(1 - zu)^2}{u}. \quad (160)$$

For later steps, use (155) with two consecutive orbit points. It expresses $d(z, x_j)$ as a bounded holomorphic factor times $x_{j-1} - x_j$, for $j \geq 1$, and the same holds for the upper orbit. Therefore both $d(z, x_j)$ and $c(z, x_j) - 1 = -(1 - zx_j)d(z, x_j)$ decrease geometrically, uniformly on compact subsets of (157).

For any consecutive range of factors,

$$\prod |c_j| \leq \prod (1 + |c_j - 1|) \leq \exp \left(\sum |c_j - 1| \right).$$

These products are bounded. Iterating an affine recurrence gives

$$w_N = w_0 \prod_{j=0}^{N-1} c_j + \sum_{k=0}^{N-1} d_k \prod_{j=k+1}^{N-1} c_j,$$

so its partial values are bounded as well. Finally $w_{j+1} - w_j = (c_j - 1)w_j + d_j$ is summable uniformly on compact subsets. This proves normal convergence of v_j, q_j, p_j and holomorphy of their limits, by the Weierstrass theorem. No division by an individual c_j is used. $\square$

Since $P(0) = 2$, the quotient in (154) is holomorphic near $(0, 1)$. Its composition with the analytic germ $R(z)$ agrees coefficientwise with the counting series. Thus F has a positive radius of convergence as a conclusion, without assuming a prior exponential bound on the counts.

24 A zero free denominator on a larger disk

A local representation alone would not exclude a pole of $1/P$ or another competing singularity on the critical circle. We now give an explicit uniform exclusion.

Proposition 24.1. *On the closed disk $|z| \leq Z = 149/1000$,*

$$|P(z)| > e^{-32/3} > 3^{-11}. \quad (161)$$

In particular P has no zeros there. It is holomorphic on a neighborhood of this disk.

Proof. Let $B_* = 4/3$; this symbol is a radius, not the generating function $B(z, x, y)$. Exact rational inequalities give

$$\rho < Z < \frac{3}{20}, \quad \frac{ZB_*}{(1 - ZB_*)^2} < B_* - 1 = \frac{1}{3}. \quad (162)$$

Thus the lower orbit stays in $|x| \leq B_*$ and, after the initial point, satisfies $|x_j - 1| < 1/3$. In particular $|x_j| > 2/3$. A derivative bound on this disk is

$$\begin{aligned} L_* &= \frac{Z(1 + ZB_*)}{(1 - ZB_*)^3} = \frac{150694875}{434163602} < \frac{7}{20}, \\ K_* &= \frac{1 + ZB_*}{(1 - Z)^2(2 - B_*)} = \frac{1798000}{724201} < \frac{5}{2}. \end{aligned} \quad (163)$$

The first difference is $x_1 - x_0 = z/(1 - z)^2$. Contraction and (147) therefore imply, for $j \geq 1$,

$$|x_{j+1} - x_j| \leq \frac{|z|L_*^j}{(1 - Z)^2}, \quad |c(z, x_j) - 1| \leq K_*L_*^j. \quad (164)$$

At zero these inequalities hold by removable continuation; indeed (155) gives $x_{j+1} - x_j = O(z^{j+1})$.

The exact rational premises also give

$$K_*L_* < \frac{7}{8}, \quad \frac{K_*L_*}{1 - L_*} < \frac{4}{3}.$$

Every factor with $j \geq 1$ is therefore nonzero, and its deviation from 1 has a summable majorant. The initial factor is

$$c(z, 1) = \frac{2 - z}{1 - z}, \quad |c(z, 1)| \geq \frac{2 - Z}{1 + Z} > 1.$$

Let $u_j = K_*L_*^j$. For $0 \leq u \leq 7/8$, $\log(1 - u) \geq -8u$, as follows by integrating $(1 - u)^{-1} \leq 8$. Hence

$$|P(z)| \geq |c(z, 1)| \prod_{j \geq 1} (1 - u_j) > \exp \left(-8 \sum_{j \geq 1} u_j \right) > e^{-32/3} > 3^{-11}.$$

The last comparison follows from $32/3 < 11$ and $e < 3$. Strict inequalities leave room to increase Z slightly; normal convergence and holomorphy also follow from Proposition 23.1. This establishes the neighborhood assertion. $\square$

For $0 < z \leq Z$, the lower real orbit is strictly increasing and stays below its attracting fixed point in $[1, 4/3]$. Thus $d_j < 0$ and $c_j > 1$, giving $P(z) > 2$; at zero $P(0) = 2$ exactly. This positivity is useful as a check, but the complex disk bound (161), rather than positivity alone, is what rules out all competing denominator poles.

25 The dominant singularity and its local expansion

From (149) and Stirling's formula, R has radius $\rho = 4/27$, and its nonnegative coefficients give

$$|R(z)| \leq R(\rho) = \frac{3}{2} \quad (|z| \leq \rho). \quad (165)$$

The branch value $3/2$ can also be obtained by following the real solution of $z = (R - 1)/R^3$ from $R = 1$ to its first stationary point. The coefficient formula ensures convergence at $z = \rho$.

The holomorphy margins $\rho < Z < 3/20$ and $3/2 < 151/100$, together with (161), show that (154) continues the original counting germ throughout $|z| < \rho$, continuously to its boundary. Consequently

$$\sum_{n \geq 0} a_n \rho^n = F(\rho) < \infty. \quad (166)$$

For example, this follows first on the positive radius and then by monotone convergence of the nonnegative coefficient sum.

For the algebraic equation $R - 1 - zR^3 = 0$, a finite nonzero critical value must satisfy

$$R - 1 - zR^3 = 0, \quad 1 - 3zR^2 = 0.$$

These equations give only $R = 3/2$, $z = 4/27$. At every other point of $|z| = \rho$, the implicit-function derivative is nonzero, so the selected branch $R(z)$ continues analytically across that point. The same then holds for F by (154). Near $z = \rho$, the algebraic root has its ordinary square-root continuation. A finite cover of the remaining compact arcs of the critical circle, plus that local slit neighborhood, gives a Δ -domain for F . This is global control of the full dominant circle, not merely a real-axis singular expansion.

Put

$$t = \sqrt{1 - z/\rho}, \quad (167)$$

with $t > 0$ when $z < \rho$ is real and close to ρ . The selected root has the convergent expansion

$$\begin{aligned} R(t) = \frac{3}{2} - \frac{\sqrt{3}}{2}t + \frac{2}{3}t^2 - \frac{35\sqrt{3}}{108}t^3 + \frac{40}{81}t^4 \\ - \frac{1001\sqrt{3}}{3888}t^5 + O(t^6). \end{aligned} \quad (168)$$

One way to verify both convergence and the branch choice is to set $R = (3/2)(1 - u)$. The defining equation becomes

$$t = u \sqrt{\frac{3 - u}{(1 - u)^3}}. \quad (169)$$

The right side has nonzero derivative $\sqrt{3}$ at zero, so it has a holomorphic inverse, with $u = t/\sqrt{3} + O(t^2)$. Expanding this inverse gives (168).

Joint holomorphy of G around $(\rho, 3/2)$ now yields a convergent power series

$$G(\rho(1 - t^2), R(t)) = \sum_{j \geq 0} b_j t^j. \quad (170)$$

The chain rule gives

$$b_1 = -\frac{\sqrt{3}}{2}G_R(\rho, 3/2), \quad C = \frac{\sqrt{3}}{4\sqrt{\pi}}G_R(\rho, 3/2). \quad (171)$$

The exact certificate in Section 26 proves $G_R(\rho, 3/2) > 0$. Thus $b_1 \neq 0$, so the singularity at ρ is genuine. It follows that the actual radius of F is ρ , and the preceding continuation proves uniqueness on that circle.

26 Exact amplitude and correction certificates

The analytic proof does not depend on floating-point fitting. We give an explicit infinite-tail bound so finite rational interval calculations can certify the constants in the theorems.

26.1 A uniform complex neighborhood

Take

$$h = 10^{-5}, \quad r = 10^{-3}, \quad q = \frac{1}{2}, \quad M_* = \frac{176}{100}, \quad Z = \frac{149}{1000}, \quad (172)$$

and work on

$$|z - \rho| \leq \rho h^2, \quad |R - 3/2| \leq r. \quad (173)$$

Here M_* is a radius for orbit arguments and is unrelated to the target size M used for inversion. All inequalities in the following derivation have exact rational premises checked by the certificate programs.

On $|x| \leq M_*$, $|z| \leq Z$, the two derivative bounds are

$$|\partial_x \phi_z(x)| \leq \frac{Z(1 + ZM_*)}{(1 - ZM_*)^3} < q, \quad |\partial_z \phi_z(x)| \leq \frac{M_*(1 + ZM_*)}{(1 - ZM_*)^3} < 6.$$

At the initial argument R^2 , whose modulus is less than $226/100$, the corresponding bounds are $11/10$ and 11 . Since $|R^2 - 9/4| \leq (3 + r)r$, the upper initial point differs from its real reference value $\phi_\rho(9/4) = 7/4$ by less than

$$\frac{11}{10}(3 + r)r + 11\rho h^2 < \frac{1}{250}.$$

The lower initial point is exactly its reference value 1.

The real reference orbits at $(z, R) = (\rho, 3/2)$ stay in $[1, 7/4]$. Comparing one complex step with a reference step gives an error recurrence $e_{j+1} \leq qe_j + 6\rho h^2$, and therefore

$$e_j \leq \frac{1}{250}q^j + 12\rho h^2 < \frac{1}{200}. \quad (174)$$

This is a self-consistent induction because $7/4 + 1/200 < M_*$. It also gives $|x_j|, |y_j| > 99/100$.

The upper reference first difference is $7/4 - 589/400 = 111/400 < 28/100$; the lower one has size $\rho/(1 - \rho)^2 < 21/100$. Incorporating the initial perturbation and then applying contraction gives, for either orbit w_j ,

$$\begin{aligned} |w_{j+1} - w_j| &< \frac{3}{10}q^j, \\ |d(z, w_j)| &< \frac{11}{5}q^j, \\ |c(z, w_j) - 1| &< 3q^j. \end{aligned} \quad (175)$$

For the second inequality use $1/|z| < 7$, $1/|w_j| < 100/99$; the third follows from $1 + ZM_*$. This neighborhood is away from $z = 0$, so no removable-limit issue arises in these estimates.

26.2 An explicit infinite tail

Every finite product of absolute c factors is less than

$$\exp\left(3 \sum_{j \geq 0} q^j\right) = e^6 < 3^6 < 2^{10} =: K_0.$$

An affine orbit with initial modulus at most 2 has all partial values bounded by

$$K_0 \left(2 + \sum_{j \geq 0} \frac{11}{5} q^j\right) < 7K_0.$$

This covers v_j, q_j, p_j , with zero additive term for p_j . By summing their increments from N onward, we obtain

$$|V - v_N|, |Q - q_N|, |P - p_N| < 2^{16} q^N =: E_N. \quad (176)$$

Indeed the increment majorant is $(3 \cdot 7K_0 + 11/5)q^j$, and its tail sum is less than $2^{16}q^N$. In the implemented certificates

$$N = 512, \quad E = E_{512} = 2^{-496}. \quad (177)$$

For the R derivative at the center, Cauchy's estimate on the disk of radius $r = 10^{-3}$ gives

$$|V_R - (v_N)_R| < 1000E. \quad (178)$$

No derivative tail of P or Q is needed because those functions do not depend on R .

26.3 Directed arithmetic and the amplitude

The real interval implementation uses integer endpoints on the lattice $10^{-120}\mathbb{Z}$. Sums are exact on this lattice; products and reciprocals round the lower endpoint downward and the upper endpoint upward. Reciprocals are taken only after checking that zero is excluded. Square roots use integer square roots and outward rounding. The constant π is enclosed using Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239) \quad (179)$$

and the alternating-series remainder after 120 and 40 terms, respectively. These operations have no floating-point input. Dual-number interval arithmetic propagates first derivatives through the same finite recurrence.

After widening finite values by (176) and the upper R derivative by (178), the quotient (154) gives

$$\begin{aligned} F(\rho) &= 1.22624332667254683644642931221778\dots, \\ P(\rho) &= 3.13161052808937020027269114323476\dots, \\ G_R(\rho, 3/2) &= 0.0007407402584171465268932549321887829\dots \end{aligned}$$

In particular the certified derivative enclosure lies strictly between 0.000740740 and 0.000740741. Equation (171) then gives (133), including its strict positivity. The displayed long decimals here are abbreviated enclosures; the complete interval endpoints are supplied in the companion.

26.4 Cauchy bounds for the Puiseux coefficients

For the first two coefficient corrections we need b_1, b_3, b_5 . To control the infinite-orbit error on a complex t disk, write (169) as

$$u = t\psi(u), \quad \psi(u) = \sqrt{\frac{(1-u)^3}{3-u}}.$$

On $|u| = 1/100$, $|\psi(u)| < 3/5$. Lagrange inversion and Cauchy's coefficient estimate give

$$|[t^k]u(t)| \leq \frac{1}{k} \left(\frac{3}{5}\right)^k 100^{k-1} \leq \frac{3}{5} 60^{k-1}.$$

Thus $u(t)$ converges for $|t| < 1/60$, and

$$|u(t)| \leq \frac{(3/5)|t|}{1 - 60|t|}. \quad (180)$$

For $|t| \leq h$, this implies $|R(t) - 3/2| < h < r$, while $|z - \rho| \leq \rho h^2$. Hence (176) applies throughout that disk.

Complex rectangular interval arithmetic encloses the finite quotient on the larger boxes containing this disk. The exact checks establish

$$\operatorname{Re} p_N > 3, \quad |v_N - q_N| < 5. \quad (181)$$

The rectangles are used only to bound these finite expressions; the infinite-tail theorem is used on the actual t disk above. Since each of the three tails is at most $E < 1/2$,

$$\left| \frac{V - Q}{P} - \frac{v_N - q_N}{p_N} \right| \leq \frac{2E}{3 - E} + \frac{5E}{3(3 - E)} < 2E.$$

Cauchy's coefficient estimate therefore bounds the error in each Taylor coefficient of the finite quotient by

$$|b_j - [t^j](v_N - q_N)/p_N| < 2Eh^{-j}. \quad (182)$$

Interval Taylor jets through degree five compute the finite coefficients; their widths are then enlarged by (182). For orientation, the resulting values are

$$\begin{aligned} b_1 &= -0.00064149988139509874897893846423296835 \dots, \\ b_3 &= 0.05852469731287832893335588719712865501 \dots, \\ b_5 &= -0.79679053577631879378776442471360871456 \dots \end{aligned}$$

Section 27 derives the formulas converting these coefficients to d_1, d_2 , and the exact intervals prove (134)–(135).

27 Transfer to every fixed order

For nonintegral α , the exact coefficient identity is

$$[z^n](1 - z/\rho)^\alpha = \rho^{-n} \frac{\Gamma(n - \alpha)}{\Gamma(-\alpha)\Gamma(n + 1)}. \quad (183)$$

Define universal polynomials $g_k(\alpha)$ by the gamma-ratio asymptotic expansion

$$\frac{\Gamma(n-\alpha)}{\Gamma(n+1)} \sim n^{-\alpha-1} \sum_{k \geq 0} g_k(\alpha) n^{-k}. \quad (184)$$

For example,

$$g_0 = 1, \quad g_1 = \frac{\alpha(\alpha+1)}{2}, \quad g_2 = \frac{\alpha(\alpha+1)(\alpha+2)(3\alpha+1)}{24}.$$

The gamma-ratio expansion, with a remainder after each fixed number of terms, follows from Stirling's expansion with remainder applied to the two gamma functions and exponentiation. It is not an assertion of convergence of the infinite series.

Proposition 27.1. *For every fixed $J \geq 0$, equation (132) holds with*

$$C_j = \sum_{m=0}^j \frac{b_{2m+1}}{\Gamma(-m-1/2)} g_{j-m}(m+1/2), \quad 0 \leq j \leq J. \quad (185)$$

Proof. Truncate the convergent local expansion (170) after degree $2J+2$. Its remainder is $O(t^{2J+3})$, equivalently $O((1-z/\rho)^{J+3/2})$, uniformly in a smaller sector at the singularity. The even powers of t in this finite truncation are polynomials in z . Their coefficients vanish once n is larger than their degrees. For the odd powers use (183), and then (184) to the order needed. The term b_{2m+1} begins at the scale $\rho^{-n} n^{-m-3/2}$, so collection at relative order n^{-j} gives precisely (185).

The Δ -domain established in Section 25 permits the standard singularity-analysis remainder transfer: a remainder $O((1-z/\rho)^\beta)$ with this nonintegral β contributes $O(\rho^{-n} n^{-\beta-1})$ here. With $\beta = J+3/2$, this is $O(\rho^{-n} n^{-J-5/2})$, exactly the remainder in (132). One can see the scale directly by deforming Cauchy's coefficient contour to two rays around the cut and a circle of radius $1/n$ around ρ : set $z = \rho(1+w/n)$, so the remainder contributes $n^{-\beta}$, the differential dz contributes n^{-1} , and z^{-n-1} provides an exponentially decaying factor along the rays. The contour away from the singularity is exponentially smaller. The standard Δ -transfer theorem formalizes this argument; see Flajolet–Sedgewick [6, Chapter VI]. $\square$

For the leading term, $\Gamma(-1/2) = -2\sqrt{\pi}$, which gives (171). Using $\Gamma(-1/2)/\Gamma(-3/2) = -3/2$ and $\Gamma(-1/2)/\Gamma(-5/2) = 15/4$, the next coefficients reduce to

$$d_1 = \frac{3}{8} - \frac{3}{2} \frac{b_3}{b_1}, \quad (186)$$

$$d_2 = \frac{25}{128} - \frac{45}{16} \frac{b_3}{b_1} + \frac{15}{4} \frac{b_5}{b_1}. \quad (187)$$

These are the formulas used by the certificates. The corrections are large: at $n = 100$, even d_1/n exceeds 1. This explains why moderate-size data can make direct extraction of the very small leading amplitude delicate. It does not, by itself, provide an error estimate at $n = 100$ or justify truncating there.

Every b_j is a derivative of the locally holomorphic composition (170). Finite jets of the orbits plus a Cauchy tail estimate analogous to (182) give an algorithm for any fixed coefficient and any prescribed rational accuracy. Higher orders may require a larger orbit length and lattice precision. Only the stated first coefficients are numerically certified in this package; the all-order result is a theorem with an algorithm, not a claim to have run certificates at every order.

28 All order inversion and integer thresholds

We now justify Theorem 20.2 without pretending that a_n is a specified continuous function of n .

28.1 Monotonicity and the logarithmic expansion

Appending the current maximum to a valid sequence produces a valid sequence of length $n + 1$: no earlier entry exceeds the appended maximum, so a new forbidden triple cannot occur. Deletion of that final entry recovers the original object. This is an injection and proves $a_{n+1} \geq a_n$ (including the empty-sequence step). Moreover (132) implies

$$\frac{a_{n+1}}{a_n} \longrightarrow \frac{27}{4} > 1,$$

so the counts are eventually strictly increasing and unbounded. The threshold $N(M)$ is consequently well-defined for every positive target M .

[write] **5 October 2026.** This paragraph proves $a_{n+1} \geq a_n$ for every n by an injection, and strict increase only for large n , from the ratio limit, which rests on Theorem 20.1. Parts I and II prove strict increase for every $n \geq 1$ by exhibiting an object outside the image of their injection; the same can be done here: Remark 28.2 at the end of this section.

For each fixed K , taking the logarithm of (132) gives

$$\log a_n = \lambda n - p \log n + \log C + \sum_{k=1}^K \frac{\ell_k}{n^k} + O_K(n^{-K-1}), \quad (188)$$

where

$$\log \left(1 + \sum_{j \geq 1} d_j w^j \right) = \sum_{k \geq 1} \ell_k w^k \quad (189)$$

is a formal identity. In particular, $\ell_1 = d_1$ and $\ell_2 = d_2 - d_1^2/2$. Set $y = \lambda n$, $L = \log M$, and $A = \log C + p \log \lambda$. The smooth asymptotic equation reads

$$L = y - p \log y + A + \sum_{k \geq 1} \ell_k \lambda^k y^{-k}. \quad (190)$$

This is an equation for asymptotic series, to be truncated at a fixed order with a controlled remainder.

28.2 The inverse polynomials

There are unique polynomials P_j such that the formal inverse has the form

$$y = L + p \log L - A + \sum_{j \geq 1} \frac{P_j(\log L)}{L^j}. \quad (191)$$

To construct them, substitute the partial expression into the right side of (190), expand the logarithm and reciprocals in L^{-1} , and cancel successive powers. At order L^{-j} the new unknown P_j appears with coefficient 1, while all other terms are already determined. This proves both existence and uniqueness recursively. The degree of P_j is at most j .

Write $q_L = p \log L - A$. The first two polynomials are

$$P_1(\log L) = pq_L - \lambda d_1, \quad (192)$$

$$P_2(\log L) = pP_1(\log L) - \frac{p}{2}q_L^2 + \lambda d_1 q_L - \lambda^2(d_2 - d_1^2/2). \quad (193)$$

For example, substitute $y = L + q_L + P_1/L + P_2/L^2$ and use

$$\log y = \log L + \frac{q_L}{L} + \frac{P_1 - q_L^2/2}{L^2} + O\left(\frac{(\log L)^3}{L^3}\right).$$

Together with $y^{-1} = L^{-1} - q_L L^{-2} + O((\log L)^2 L^{-3})$ and $y^{-2} = L^{-2} + O((\log L) L^{-3})$, this yields (192)–(193). Dividing (191) by λ and retaining P_1 gives (136).

28.3 A rigorous discrete localization

For $J \geq 0$ let

$$\nu_J(L) = \frac{1}{\lambda} \left[L + p \log L - A + \sum_{j=1}^J \frac{P_j(\log L)}{L^j} \right]. \quad (194)$$

In particular, even the $J = 0$ truncation retains the constant $-A$. Keeping only L/λ , or omitting another nonvanishing term, would not justify an error interval tending to zero.

Proposition 28.1. *For every fixed $J \geq 0$ there exist $B_J > 0$ and L_J such that, for all $L \geq L_J$, setting*

$$\varepsilon_J(L) = B_J \frac{(1 + \log L)^{J+1}}{L^{J+1}},$$

one has

$$[\nu_J(L) - \varepsilon_J(L)] \leq N(e^L) \leq [\nu_J(L) + \varepsilon_J(L)]. \quad (195)$$

Proof. Choose $K \geq J + 1$ in (188), and let

$$g_K(x) = \lambda x - p \log x + \log C + \sum_{k=1}^K \ell_k x^{-k}, \quad x > 0.$$

There are $B > 0, n_0$ such that for integers $n \geq n_0$,

$$g_K(n) - Bn^{-K-1} \leq \log a_n \leq g_K(n) + Bn^{-K-1}.$$

Both bounding smooth functions $g_-(x) = g_K(x) - Bx^{-K-1}$ and $g_+(x) = g_K(x) + Bx^{-K-1}$ are strictly increasing for sufficiently large x , because their derivatives tend to $\lambda > 0$. For large L , let their unique large inverse roots be $x_-(L) = g_-^{-1}(L)$ and $x_+(L) = g_+^{-1}(L)$; the ordering is $x_+(L) \leq x_-(L)$. If an integer $n < x_+(L)$, its upper bound is less than L , so $a_n < e^L$. If $n \geq x_-(L)$, its lower bound is at least L , so $a_n \geq e^L$. Early integers are irrelevant once L exceeds their finitely many logarithmic counts. Therefore

$$[x_+(L)] \leq N(e^L) \leq [x_-(L)].$$

The substitution construction of (191) gives a residual at $\nu_J(L)$ of $O((1 + \log L)^{J+1}/L^{J+1})$ in each smooth equation. Their derivatives are bounded below by, say, $\lambda/2$ there. The mean value theorem thus places both inverse roots within $\varepsilon_J(L)$ of $\nu_J(L)$ after increasing B_J . Taking ceilings proves the result. $\square$

The constants B_J, L_J are existence constants here. The exact coefficient intervals do not automatically make these threshold bounds numerically effective at a specified moderate M . An explicit certified value of $N(M)$ would require either a quantified asymptotic remainder and threshold or exact enumeration near the candidate. The theorem nevertheless gives rigorous all-order localization as $M \rightarrow \infty$, and in particular $N(M) = \nu_1(\log M) + O(1)$.

[write] **5 October 2026.** Proposition 28.1 is the bracket of Part I's Theorem 7.1 and Part II's Theorem 17.1 for A279558, in other coordinates: those keep the Lambert centre u exact and expand in powers of $1/u$, with error $O(u^{-K-1})$; this Part expands the centre itself in powers of $1/L$ with coefficients polynomial in $\log L$, as in (191), which costs the factor $(1 + \log L)^{J+1}$ in ε_J . Its expansion, like Part II's (128), has the form of the direct polynomial-logarithmic reversion `p0:thm:flattening` of the transseries volume [9] (there $L = \log(y/C)$ and the coefficients are polynomials in $H = \log(L/a)$; here $L = \log M$ and they are polynomials in $\log L$, a re-expansion), whose remainder $O((1 + H)^{N+1}/L^{N+1})$ is the factor $(1 + \log L)^{J+1}$ of ε_J , as `p0:rem:accuracy-claim` records in comparing the two forms; the bracket is an instance of the staircase results as in the note after Theorem 7.1. (Dated note, 5 October 2026: corrected after the independent check. This sentence first read "It is an instance of the transseries volume's Lambert-core and staircase results in the same way (note after Theorem 7.1)", which names no theorem of the volume for the expansion in powers of $1/L$.)

Remark 28.2 ([write], 5 October 2026). For every $n \geq 1$, $a_{n+1} \geq a_n + 1$; in particular $a_{n+1} > a_n$ for $n \geq 1$, while $a_1 = a_0 = 1$.

Proof. Let $n \geq 1$. For e in the class of length n , with maximum h , appending h gives a sequence in the class (Section 28: a forbidden triple ending at the new entry would need an earlier entry $e_j > h$), and deleting the last entry recovers e . So appending the current maximum is an injection from length n to length $n + 1$, and every sequence in its image has last entry equal to the maximum of its first n entries. Now consider $f = (0, \dots, 0, n)$ of length $n + 1$. It is an inversion sequence, since $n < n + 1$. It avoids (129): a triple $i < j < k$ needs $e_j > e_k$, but $e_j = 0$ for every $j \leq n$, and $e_k \geq 0$. Its last entry $n \geq 1$ differs from the maximum 0 of its first n entries, so f is not in the image. Hence $a_{n+1} \geq a_n + 1$. (A brute-force count at the write gives 1, 1, 2, 5, 15, 52, 200, 830, 3654, 16869, 80963 for $n = 0, \dots, 10$, the displayed terms of A279558.) $\square$

29 Reproducibility and what the checks establish

The companion is designed for offline use with standard-library Python. Its exact computations supplement the infinite analytic proof; finite enumeration alone does not prove the asymptotic theorem. The editable source of this report and a deterministic PDF build are included.

The exact model checks compare direct forbidden-triple avoidance against the transformed tree through length 9, including the full $(a, b, S/T)$ distributions, not just totals. They match all 26 terms displayed on the cited OEIS page and compute exact terms through length 150. A separate bivariate calculation checks the cleared scalar equation through z^{40} . Formal-series checks verify the algebraic anchor through z^{64} and the finite-orbit agreement (156) for $m = 0, \dots, 30$, through the predicted degree $m + 1$. The supplied 151-term table is generated from the tree; it is not represented as a downloaded OEIS b-file.

The analytic certificate computations check rational premises for the contracting disks and the global zero-free product, and compute outward-rounded enclosures for the finite orbits, the amplitude and the Taylor coefficients. The infinite tails come from the proved estimates (176), (178) and (182). The companion reimplements these finite computations with active guards and checks its fixed

fixtures; this is a computational cross-check, not a proof-assistant formalization of the complex analysis.

The top-level `README.md` documents the commands, dependency requirements, immutable reader inventory, exact tested ranges, and negative tests. In brief, from an extracted copy use

```
python3 reproduce.py
bash build.sh
python3 repack.py /path/outside/package/report122.zip
```

The authoritative companion is under `checks/`. Cleaned versions of the original certificate and model programs are under `scripts/`; they are auxiliary and explicitly reject Python optimization modes rather than silently disabling assertions. All generated outputs belong outside the immutable reader tree. The package integrity verifier detects additions, removals and byte changes in its closed inventory. The reproducibility wrapper uses temporary output directories and leaves the source tree unchanged. Identical PDF bytes are required from two clean builds in the recorded TeX environment; other TeX distributions may give different typesetting bytes without changing the mathematics.

[write] **5 October 2026.** The commands above, `checks/`, `scripts/` and “the top-level `README.md`” refer to the delivered package. In this repository its files are shipped under the prefix `122-anchor-` in `code/` and `data/`, with `122-anchor-checks-README.md`; the delivered `README.md`, the PDF, `CHECKSUMS.sha256` and `checks/inventory.json` are not shipped. Because both inventories are closed and the scripts read the delivered layout, the checks are rerun from the archive; on Windows the outputs are written with CRLF line ends, so the byte comparison of `reproduce.py` fails there although the content agrees (see the report’s `README`).

30 Further questions and limitations

The orbit solution settles the leading equivalent and the regular all-fixed-order scale for this class. Several natural questions remain.

1. **Effective remainder constants.** The present argument gives O_J remainders and existence thresholds. Explicit contour bounds could turn them into numerical inequalities for all $n \geq n_0(J)$, making the inverse thresholds directly executable.
2. **Higher correction coefficients.** The jet and Cauchy method computes any prescribed fixed order. An efficient implementation could optimize the choice of disk radius, orbit length and lattice precision for large orders.
3. **The large correction regime.** The first relative corrections are large despite the small leading amplitude. Quantifying optimal truncation or obtaining uniform finite- n bounds would explain the crossover more sharply than leading asymptotics alone.
4. **Global analytic structure.** The disk $|z| \leq 149/1000$ suffices for dominant-circle control. It does not describe all farther singularities, zeros of P , or the maximal analytic continuation of the orbit quotient.
5. **Algebraicity and D-finiteness.** A representation using an algebraic anchor and infinite rational orbits does not decide either property. No conclusion about them is drawn here.

6. **Refined statistics.** The catalytic variables retain maximum and premaximum information. Extending the orbit analysis to marked observables may yield distributional results, but those are not proved by the univariate argument in this report.

The numerical amplitude was known before this proof, and the published source already proposed the regular scale. The proof, all-order transfer and honest discrete inversion are the objects to assess. A limited source and repository search found no matching proof during preparation; it is not a comprehensive priority certification. No publication or external repository modification is part of the accompanying reproducibility work.

[write] **5 October 2026.** The “repository search” above is the delivering session’s. Before this report, nothing in ProveIt treated A279558 or any inversion-sequence class of this kind (searched at placement).

30.1 Further questions and research

[write] (5 October 2026.) Under Vladimir’s standing rule of 4 October 2026, this subsection collects every claim of Part III that is stated without proof or left open, with its source, the sketch offered and what is missing. Nothing in this Part was found to be wrong, and no claim was refuted; the one statement proved only in a weaker form, strict monotonicity, is completed by Remark 28.2.

1. *Effective remainder constants* (item 1 above; the paragraph after Theorem 20.1, “We do not claim an explicit numerical threshold n_0 ”; the paragraph after Proposition 28.1). Sketch: explicit contour bounds. Missing: an explicit Δ -domain with bounds, hence $n_0(J)$ and certified B_J, L_J .
2. *Higher correction coefficients* (item 2; Section 27: “a theorem with an algorithm”). The algorithm for any fixed coefficient (finite orbit jets plus a Cauchy tail analogous to (182)) is described, but it has been run only for b_1, b_3, b_5 , that is for C, d_1, d_2 ; the choice of radius, orbit length and lattice precision for higher orders is open.
3. *The large-correction regime* (item 3). The first corrections are large ($d_1/n > 1$ at $n = 100$). Open: uniform finite- n bounds, optimal truncation, and an explanation of the size of d_1, d_2 .
4. *Global analytic structure* (item 4). Farther singularities, zeros of P beyond $|z| \leq 149/1000$ and the maximal continuation of the orbit quotient are not described.
5. *Algebraicity and D -finiteness* (item 5). Source: Britt and Beaton “do not expect” the class to be algebraic (their Section 2.4). Neither property is decided by the anchor-and-orbit representation.
6. *Refined statistics* (item 6): distributional results for the maximum and premaximum carried by the catalytic variables.
7. *Earlier rigorous proofs.* The search recorded above was limited; the wider literature was not searched at placement or at the write (as Part I, item 6).

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of the write, made by the intake after the write (37c5ebe97), reread the seven items of the write that carry mathematics or citations: Remark 10.2 and the transfer to class 1953B; Remark 28.2; the note identifying Theorem 20.2 with Proposition 28.1 at $J = 1, B = 4B_1$; the note identifying

(128) with (136) and its error term with the proof of Proposition 28.1; the sentences comparing Kotěšovec’s amplitudes with the certified intervals; the citations of the transseries volume [9]; and the transfer formulas for c_1, c_2 in “The common pipeline”. It found every mathematical claim valid, with no counterexample and no gap in any proof chain. One sentence of the abstract and one row of “What is proved, and where” said more than the mathematics: Part III’s inverse contains no Lambert W_{-1} (Theorem 20.2 and Proposition 28.1 expand in powers of $1/L$ with coefficients polynomial in $\log L$), and its error is $O((1 + \log L)^{J+1}/L^{J+1})$, not $O(u^{-K-1})$. Both are corrected where they stand, with dated notes keeping the first wording. The theorem of the volume that Part III’s expansion and Part II’s (128) instantiate, `p0:thm:flattening`, was not cited; after its label and statement, and those of `p0:rem:accuracy-claim`, were confirmed in the volume, it is cited in the front matter, in the note after Proposition 28.1 (whose first wording a dated note keeps) and in the bibliography. The hedges on Martinez and Savage (“not checked here”, “not consulted”, “not re-checked here beyond $n \leq 10$ ”) are replaced by the precise citation, their Theorem 60, read in arXiv:1609.08106v2, in Remark 10.2, in item 6 of Section 19.1 and in the bibliography, each with a dated note; the statement that no bijection carrying Part II’s generating tree to class 1953B is given stands. Two precisions were made in notes: the note after Theorem 20.2 writes ε_1^P for the error of Proposition 28.1 at $J = 1$, which it had also called ε_1 , the letter of the theorem’s own error; and the staircase of `p0:def:three-inverses` is matched to $N(Y)$ for $Y > 1$ with $n_1 = 1$, since $a_0 = a_1 = 1$. No other statement or number changed.

The check used neither the delivered programs nor the write’s. It counted classes 214, 1509, 1953A, 1953B and 830 by brute force from the forbidden-relation definitions for $n \leq 10$, and by its own generating-tree recurrences, derived from the definitions and not from the report: class 214 over its two largest entries counted with multiplicity, class 1509 over its maximum and a threshold, class 1953A over its two largest entries, class 1953B over its maximum, second distinct value and a threshold, and class 830 over its maximum, its premaximum and a flag, which reproduces independently Part III’s rule that a sequence of the class ends on its maximum or premaximum, and its S/T split. Each recurrence equals the brute force for $n \leq 10$ and the OEIS b-file (fetched 5 October 2026) for $n \leq 60$ (A279544, A279567, A279569 and A279558), and the OEIS definitions agree with the report’s pattern sets. The recurrence for class 1953B equals that for class 1953A and the A279569 b-file for every $n \leq 80$, and class 830 satisfies $a_{n+1} - a_n \geq 1$ for $1 \leq n < 60$. In Remark 28.2 it checked the choice of injection: under the other natural injection, appending n , which also stays in class 830, the extra object $(0, \dots, 0, n)$ would lie in the image, but the remark appends the maximum, and there it does not. With SymPy it confirmed P_1 and P_2 of (192)–(193), since substituting $y = L + q_L + P_1/L + P_2/L^2$ in (190) leaves zero residual at the orders L^0, L^{-1} and L^{-2} , and that $B(L) = q_L$ and $pB(L) - \lambda d_1 = P_1(\log L)$ once $A = \log C + p \log \lambda$ is substituted. On the b-files, with the certified C, d_1, d_2 , the scaled error $(\nu_1(\log a_n) - n)L^2/(\log L)^2$ stays bounded: between -246 and -181 for A279558 at $100 \leq n \leq 1000$, and at most 25 in absolute value for A279569 at $100 \leq n \leq 400$. At 80 digits it compared Kotěšovec’s amplitudes with the certified intervals: the values for A279544 and A279567 lie inside (24) and inside both enclosures (56) and (63) of $\widehat{C}$; the value for A279569 lies inside (77) and equals both endpoints of (115) truncated to 31 decimals; and Kotěšovec’s c for A279558 equals both endpoints of $C\sqrt{\pi}$ truncated to 39 decimals, the OEIS formula $a_n \sim c(27/4)^n/(\sqrt{\pi}n^{3/2})$ confirming the normalization. It re-derived c_1 and c_2 of “The common pipeline” from the expansion of $[z^n](1 - z)^\alpha$ at $\alpha = \frac{1}{2}$ and $\frac{3}{2}$ and the ratios $\Gamma(-\frac{1}{2})/\Gamma(-\frac{3}{2}) = -\frac{3}{2}$ and $\Gamma(-\frac{1}{2})/\Gamma(-\frac{5}{2}) = \frac{15}{4}$, and checked the citations of the volume against its text: the branch rule of `p0:thm:lambert-core` gives (64), the Bell recurrence of `p0:thm:lambert-centered` reproduces (71) and (126), and `p0:cor:forward-to-inverse`, `p0:def:three-inverses` and `p0:thm:staircase` apply as stated. This was a careful reading with numerical tests, not a formal verification.

References

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- [6] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009, Chapter VI (singularity analysis and transfer theorems). <https://ac.cs.princeton.edu/home/>.
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- [8] [write] M. A. Martinez and C. D. Savage, Patterns in inversion sequences II: inversion sequences avoiding triples of relations, *Journal of Integer Sequences* **21** (2018), Article 18.2.2. Preprint arXiv:1609.08106v2 (8 January 2018), <https://arxiv.org/abs/1609.08106v2>: its Theorem 60 (Section 3.2.1) proves the Wilf-equivalence of classes 1953A and 1953B. Read at the independent check of 5 October 2026, in the arXiv version only; the journal version, whose numbering may differ, was not consulted. (Dated note: this entry first read “Cited from Britt–Beaton’s bibliography for the Wilf-equivalence of classes 1953A and 1953B; not consulted at this write.”)
- [9] [write] ProveIt, *Transseries and inversion*, the transseries-and-inversion volume (p0:thm:lambert-core, p0:thm:lambert-centered, p0:def:three-inverses, p0:thm:staircase, p0:cor:forward-to-inverse; added 5 October 2026 after the independent check: p0:thm:flattening, p0:rem:accuracy-claim), *Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex*.